# When to Alert, When to Defer: Conformal Selective Triage for ICU Mortality in Sepsis-Associated Acute Kidney Injury

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**Abstract**

**Background.** Sepsis-associated acute kidney injury (SA-AKI) carries high in-hospital mortality, yet machine learning studies for this population typically report a single discrimination metric and overstate deployment readiness. Clinicians require not only a risk score but also an explicit signal indicating whether the model's prediction is reliable enough to act upon.

**Methods.** A total of 10,036 ICU stays (9,002 unique subjects; 26.7% mortality) from the Medical Information Mart for Intensive Care IV (MIMIC-IV) version 3.1 were analysed. A total of 162 features were extracted from the first 24 hours of ICU admission using nine statistical aggregations over time-varying measurements. Conventional baselines (logistic regression, XGBoost, LightGBM) were evaluated under subject-grouped train/validation/test splits repeated across 21 random seeds. Mondrian conformal prediction was applied to convert calibrated risk scores into set-valued triage decisions (*Alert* $\{1\}$, *Clear* $\{0\}$, or *Defer* $\{0, 1\}$) with finite-sample coverage guarantees. A disagreement-based selective-triage comparator was included as an ablation.

**Results.** The best conventional baseline was XGBoost with isotonic calibration (mean AUROC = 0.768; recall at PPV $\geq$ 0.50: 0.608). Even a leak-augmented ceiling experiment that included time-to-event as a feature reached only AUROC = 0.810, confirming bounded discrimination. Single-model conformal triage at $\alpha = 0.05$ achieved coverage of 0.952, alert positive predictive value (PPV) of 0.643, and clear negative predictive value (NPV) of 0.948. Multi-model union consensus concentrated decisions into a smaller actionable region with alert PPV of 0.729 and clear NPV of 0.962. On decision-curve analysis the selected continuous model achieved net benefit 0.133 at a threshold probability of 0.20, compared with 0.082 for APACHE-III and 0.082 for SOFA. Under simulated missingness shift, conformal coverage remained stable while fixed-threshold policies lost recall.

**Conclusions.** The principal contribution is not a new discrimination leader but an *Alert / Defer / Clear* triage framework with distribution-free coverage guarantees. For this SA-AKI cohort the scientific opportunity lies in safe decision-making under bounded predictive signal rather than incremental AUROC improvement.

**Keywords:** sepsis-associated acute kidney injury, conformal prediction, selective triage, ICU mortality, MIMIC-IV, uncertainty quantification

## 1 Introduction

Sepsis-associated acute kidney injury (SA-AKI) couples the systemic inflammatory cascade of sepsis with acute renal failure, producing a clinical phenotype that carries substantially higher intensive care unit (ICU) mortality than either condition alone [Poston and Koyner, 2019, Kellum et al., 2021]. Reported in-hospital mortality rates for SA-AKI range from 25% to 45% depending on cohort definitions and severity staging [Hoste et al., 2015, Bagshaw et al., 2008], and survivors face elevated risks of chronic kidney disease progression, prolonged hospitalisation, and increased long-term healthcare utilisation [See et al., 2019]. Early identification of patients at highest risk of death is therefore a prerequisite for timely escalation of care, whether that involves intensifying haemodynamic support, initiating renal replacement therapy, or triggering goals-of-care discussions [Zarbock et al., 2023].

Despite the clinical importance of this prediction task, the machine learning literature on SA-AKI mortality has followed a familiar pattern: studies report a single area under the receiver operating characteristic curve (AUROC), declare the model "promising," and do not address the operational question that matters at the bedside: *when should the model's output actually change a clinical decision?* This gap between discrimination and deployment readiness is not unique to SA-AKI. Across critical care prediction tasks, models are rarely accompanied by explicit uncertainty quantification, calibration audits, or decision-theoretic evaluations that would be necessary before a risk score could safely enter a clinical workflow [Van Calster et al., 2019, Steyerberg et al., 2010]. The result is a growing inventory of models with similar AUROCs and no clear path to clinical integration.

The present study begins from a clear empirical observation: discrimination in this cohort is bounded. A total of 10,036 ICU stays from MIMIC-IV v3.1 were analysed, and it was found that the strongest conventional models (logistic regression, XGBoost, LightGBM, and CatBoost) occupy a narrow performance band with mean AUROCs between 0.765 and 0.768 under subject-grouped evaluation repeated across 21 random seeds. Grouped Optuna hyperparameter optimisation (40 trials) and the addition of CatBoost as a diversity comparator did not materially widen this band. Even a deliberately leak-prone ceiling experiment that included time-to-event as a predictor reached only AUROC = 0.810. This finding reframes the research question: the central challenge is not how to extract another 0.01 of AUROC from the same first-24-hour feature set, but how to build a clinically legible triage policy that behaves safely under bounded signal.

Two uncertainty-aware selective-triage strategies are therefore compared. The first is a *disagreement-based* approach, included as an ablation, in which divergence between a physiology-only model and a process-enriched full-feature model serves as a heuristic defer signal. The second, and primary contribution, is *conformal selective triage*. Mondrian conformal prediction [Vovk et al., 2005] converts calibrated probabilistic risk scores into set-valued decisions with finite-sample coverage guarantees. For binary mortality prediction the prediction set takes one of three clinically interpretable forms: $\{1\}$ (*Alert*, indicating confidence that the patient is high-risk), $\{0\}$ (*Clear*, indicating confidence that the patient is lower-risk), or $\{0, 1\}$ (*Defer*, indicating that the model's uncertainty is too large for safe automation and the case should be reviewed by a clinician). This three-way interface maps directly onto ICU triage workflows and provides an explicit abstention channel that fixed-threshold classifiers lack.

This manuscript makes three contributions. First, the prediction ceiling for first-24-hour SA-AKI mortality risk is documented under a rigorous subject-grouped evaluation design, establishing that incremental model engineering yields diminishing returns in this cohort. Second, conformal selective triage is introduced as the primary decision framework, and it is demonstrated that it provides explicit, safety-oriented operating points characterised by coverage, alert positive predictive value (PPV), and clear negative predictive value (NPV), with multi-model consensus further concentrating decisions into a smaller, more reliable actionable region. Third, it is shown that conformal deferral degrades more gracefully than fixed thresholds under simulated distribution shift (random missingness injection and process-feature dropout), offering a more realistic deployment

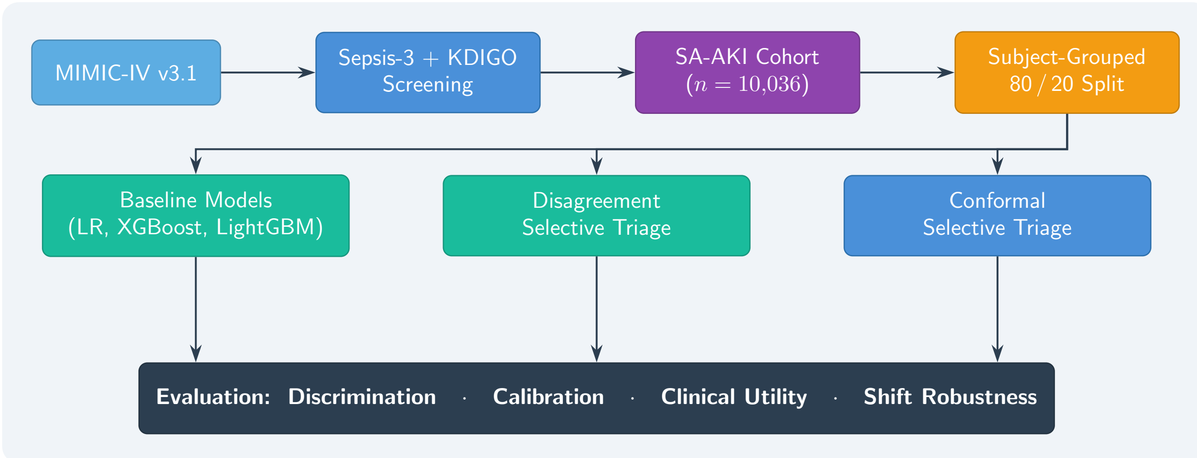


**Figure 1:** End-to-end study workflow. The raw SA-AKI cohort is derived from MIMIC-IV v3.1 via Sepsis-3 and KDIGO criteria, audited against a data contract, and split into subject-grouped partitions. Three evaluation tracks (conventional baselines, disagreement-based selective triage (ablation), and Mondrian conformal selective triage (primary)) feed into a unified manuscript asset pipeline that produces all tables, figures, and decision-curve analyses reported in this paper.

boundary for ICU mortality risk models. The goal is not to claim bedside readiness but to present a clinically grounded informatics study that defines when a model should and should not be trusted.

## 2 Methods

Figure 1 summarises the end-to-end pipeline from cohort construction through evaluation. Each subsection below corresponds to a stage in this workflow.

### 2.1 Data Source

This retrospective study used the Medical Information Mart for Intensive Care IV (MIMIC-IV) version 3.1 [Johnson et al., 2023, Goldberger et al., 2000], a freely available critical care database hosted on PhysioNet that contains de-identified electronic health records from Beth Israel Deaconess Medical Center (BIDMC), Boston, Massachusetts, spanning admissions from 2008 to 2019. MIMIC-IV integrates data from three modules: `hosp` (hospital-wide administrative and laboratory data), `icu` (high-resolution bedside charted events from MetaVision), and `ed` (emergency department encounters).

Source tables were ingested into a PostgreSQL analytical warehouse to enable reproducible, SQL-driven cohort construction. Clinical concept identification relied on an ontology-driven pipeline rather than brittle hard-coded drug lists. Specifically, systemic antibiotic and vasopressor flags, required for Sepsis-3 labelling and feature construction respectively, were derived by mapping MIMIC-IV medication entries through a crosswalk linking SNOMED CT concepts to Unified Medical Language System (UMLS) Concept Unique Identifiers (CUIs), then resolving to RxNorm ingredient-level value sets, with the Anatomical Therapeutic Chemical (ATC) classification used as an orthogonal cross-reference and coverage check. This approach ensures that the drug identification layer generalises across formulary variations and is robust to synonym differences in medication naming.

## 2.2 Cohort Construction

The analytic cohort was constructed through a series of inclusion and exclusion criteria applied sequentially to all ICU stays in MIMIC-IV v3.1. The pipeline follows a CONSORT-style reduction and is summarised below.

**Inclusion criteria.** Three inclusion criteria were applied sequentially: (1) adult patients aged $\geq 18$ years at ICU admission; (2) only the index ICU admission per hospital encounter, to avoid within-subject correlation from repeated stays; and (3) a minimum ICU length of stay of $\geq 24$ hours, ensuring that the full T0–T24 feature extraction window was available for every patient.

**Sepsis-3 identification.** Sepsis was defined according to the Third International Consensus Definitions [Singer et al., 2016]. A patient was classified as having *suspected infection* if a body-fluid culture was ordered and a systemic antibiotic was administered within a clinically plausible window (culture order within $\pm 72$ hours of antibiotic start, with the antibiotic initiated within 24 hours before or 72 hours after the culture). Sepsis was confirmed when suspected infection co-occurred with an acute increase in organ dysfunction, operationalised as a change in the Sequential Organ Failure Assessment (SOFA) score of $\Delta\text{SOFA} \geq 2$ points relative to a pre-infection baseline. The sepsis onset time, $t_{\text{sepsis}}$, was defined as the beginning of the first qualifying ICU calendar day on which both conditions were met.

**KDIGO acute kidney injury staging.** AKI was identified using the Kidney Disease: Improving Global Outcomes (KDIGO) 2012 criteria [Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012], which define AKI by any one of three triggers:

- **Absolute creatinine rise:** serum creatinine (SCr) increase $\geq 0.3$ mg/dL within any 48-hour window.
- **Relative creatinine rise:** SCr $\geq 1.5\times$ baseline within 7 days. Baseline SCr was established hierarchically: the lowest outpatient value in the 7–365 days before admission was preferred; when unavailable, the admission SCr was used.
- **Urine output criterion:** urine output $\leq 0.5$ mL/kg/h sustained for $\geq 6$ consecutive hours, computed via a sliding-window sessionisation algorithm applied to hourly charted urine volumes normalised by admission body weight.

The AKI onset time, $t_{\text{aki}}$, was defined as the earliest time at which any of the three criteria was first satisfied.

**SA-AKI temporal linkage.** A patient was classified as having SA-AKI if AKI onset fell within a $\pm 48$-hour window around sepsis onset:

$$t_{\text{sepsis}} - 48\text{ h} \;\leq\; t_{\text{aki}} \;\leq\; t_{\text{sepsis}} + 48\text{ h}. \tag{1}$$

This window captures both AKI that develops as a consequence of sepsis and AKI that is already evolving at the time sepsis is recognised, consistent with contemporary definitions of sepsis-associated organ dysfunction [Kellum et al., 2021].

**Exclusion criteria.** Two additional exclusions were applied to the SA-AKI cohort. First, patients with pre-existing end-stage renal disease (ESRD), identified via ICD-9/10 diagnosis codes or documented chronic dialysis, were excluded because their renal trajectory is fundamentally different from acute injury. Second, patients who received renal replacement therapy within the first 24 hours of ICU admission were excluded to avoid confounding the T0–T24 feature window with an intervention that simultaneously reflects severity and alters downstream laboratory values.

**Final cohort.** After applying all criteria, the analytic cohort comprised **10,036** ICU stays from **9,002** unique subjects, with an in-hospital mortality prevalence of **26.7%**. The primary outcome was in-hospital death (the binary in-hospital mortality indicator, coded as 1); the secondary time-to-event variable was hours from ICU admission to death or discharge (the time-to-event variable, measured in hours from ICU admission to death or censoring), with patients discharged alive treated as right-censored.

## 2.3 Feature Engineering

All predictive features were restricted to measurements recorded within the first 24 hours of ICU admission (T0–T24) to ensure clinical actionability and prevent temporal leakage. A scripted audit verified that no variable encoding information after T24 (including total ICU length of stay and whole-stay aggregates) was included in the predictor set.

**Time-varying features.** For each time-varying clinical measurement (vital signs, laboratory values, fluid balance, urine output), nine statistical aggregations were computed over the T0–T24 window:

1. **First** and **last** recorded values (capturing initial presentation and 24-hour trajectory endpoint);
2. **Median** (robust central tendency);
3. **Interquartile range (IQR)** and **range** (max − min) (variability);
4. **Delta** (last − first) (directional change);
5. **Area under the curve (AUC)** via trapezoidal integration (cumulative exposure);
6. **Slope** from ordinary least-squares regression against time (rate of change);
7. **Count** of non-missing observations (measurement intensity).

This nine-aggregation scheme was applied to vital signs (heart rate, mean arterial pressure, systolic and diastolic blood pressure, respiratory rate, $SpO_2$, temperature), laboratory values (serum creatinine, blood urea nitrogen, lactate, bilirubin, platelets, white blood cell count, haemoglobin, bicarbonate, potassium, sodium, glucose, arterial pH, $PaO_2$, $PaCO_2$), cumulative fluid balance, and hourly urine output.

**Static and severity features.** Static features included age, sex, admission body weight, and 17 Charlson comorbidity index components (diabetes, congestive heart failure, chronic pulmonary disease, renal disease, liver disease, malignancy, among others). Severity scores comprised the SOFA score at 24 hours (with individual organ sub-scores) and the APACHE-III score computed at ICU admission.

**Therapy indicators.** Binary and duration-based indicators captured whether the patient received mechanical ventilation, vasopressor infusion, or renal replacement therapy during the T0–T24 window, along with the cumulative duration of each intervention.

**Final feature set.** The resulting feature matrix contained **162** features per ICU stay. Missingness was handled in three steps: (i) Little's MCAR test was performed on the training partition and rejected the null hypothesis of completely random missingness; (ii) binary missingness indicators were created for each feature with missing values, and only those indicators significantly associated with mortality ($\chi^2$ test, $p < 0.05$) were retained as additional predictors; (iii) remaining missing values were imputed with the training-set median. Highly correlated feature pairs (Pearson $r > 0.99$) were deduplicated, and all numerical features were standardised using parameters fitted exclusively on the training partition.

## 2.4 Models

**Conventional baselines.** Three families of supervised classifiers were trained for binary mortality prediction. $\ell_2$-regularised logistic regression (LR) served as a transparent linear baseline. XGBoost [Chen and Guestrin, 2016], employing gradient-boosted decision trees with histogram-based splitting, and LightGBM [Ke et al., 2017], employing leaf-wise gradient boosting with exclusive feature bundling, provided two complementary tree-ensemble approaches. CatBoost [Prokhorenkova et al., 2018] was included as a diversity comparator to test whether a distinct gradient-boosting family (ordered boosting with symmetric trees) materially changed the model-selection narrative.

**Hyperparameter optimisation.** For LightGBM, a grouped Optuna search [Akiba et al., 2019] was conducted over 40 trials, optimising a composite objective that combined AUROC, recall at PPV $\geq 0.50$, and Brier score on the grouped validation split. The tuned configuration was frozen and carried into the repeated grouped benchmark. XGBoost and logistic regression used default or lightly tuned hyperparameters to preserve interpretability of the model-comparison story.

**Calibration.** Post-hoc calibration was applied to all models. Both sigmoid (Platt scaling) and isotonic regression recalibration were benchmarked on the held-out grouped validation split, and the variant with lower expected calibration error (ECE) was selected for each model. Calibration quality was assessed via reliability diagrams, calibration intercept, and calibration slope.

**Score-only comparators.** SOFA and APACHE-III scores were evaluated as single-feature logistic baselines under the same grouped protocol, providing clinical-score anchors for decision-curve analysis.

## 2.5 Conformal Selective Triage

The primary methodological contribution is the application of Mondrian conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2021] to convert calibrated risk scores into set-valued triage decisions with finite-sample coverage guarantees.

Mondrian conformal prediction was selected over standard (label-unconditional) conformal prediction because it provides class-conditional marginal coverage, which is the stronger guarantee for imbalanced classification problems. With a mortality prevalence of 26.7%, standard conformal prediction could satisfy its marginal coverage guarantee by achieving high coverage on the majority class (survivors) while under-covering the minority class (deaths). Mondrian conformal prediction avoids this failure mode by computing separate nonconformity thresholds for each class.

**Formulation.** Let $f : \mathcal{X} \to [0,1]$ denote a calibrated risk model and let $\alpha \in (0,1)$ be the desired miscoverage rate. The data available for conformal inference are partitioned into three disjoint subsets:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \,\dot\cup\, \mathcal{D}_{\text{cal}} \,\dot\cup\, \mathcal{D}_{\text{test}}, \tag{2}$$

where $\mathcal{D}_{\text{train}}$ is used to fit $f$, $\mathcal{D}_{\text{cal}}$ is used to estimate class-conditional nonconformity thresholds, and $\mathcal{D}_{\text{test}}$ receives set-valued predictions.

Post-hoc calibration was fitted via three-fold cross-validation internal to $\mathcal{D}_{\text{train}}$ using `CalibratedClassifierCV`; the calibration map was therefore learned entirely within the training partition. The conformal calibration set $\mathcal{D}_{\text{cal}}$ was a separate held-out split carved from the training data (approximately 20% of the training partition) and was not used in any step of model fitting or recalibration. This separation ensures that the nonconformity scores computed on $\mathcal{D}_{\text{cal}}$ satisfy the exchangeability requirement with respect to $\mathcal{D}_{\text{test}}$, preserving the finite-sample coverage guarantee.

For each calibration example $(x_i, y_i) \in \mathcal{D}_{\text{cal}}$, the nonconformity score for class $y \in \{0,1\}$ is defined as

$$s_i^{(y)} = 1 - \hat{p}_y(x_i), \tag{3}$$

where $\hat{p}_y(x_i)$ denotes the model's estimated probability for class $y$. Mondrian conformal prediction computes class-conditional quantiles: for each class $y$, the threshold $\hat{q}_y$ is the $\lceil (1-\alpha)(|\mathcal{D}_{\text{cal}}^{(y)}| + 1)\rceil / |\mathcal{D}_{\text{cal}}^{(y)}|$ quantile of the nonconformity scores $\{s_i^{(y)} : y_i = y\}$.

**Set-valued predictions.** For a new test example $x_{\text{new}}$, the prediction set is

$$C(x_{\text{new}}) = \bigl\{ y \in \{0,1\} : s_{\text{new}}^{(y)} \le \hat{q}_y \bigr\}. \tag{4}$$

This yields three clinically interpretable outcomes:

- $C(x) = \{1\}$: **Alert**, indicating confidence that the patient is high-risk; escalate monitoring or intervention.
- $C(x) = \{0\}$: **Clear**, indicating confidence that the patient is lower-risk; standard care pathway.
- $C(x) = \{0,1\}$: **Defer**, indicating that the model's uncertainty is too large for safe automation; the case requires clinician review.

In principle, the prediction set may also be empty ($C(x) = \emptyset$) when neither class exceeds its nonconformity threshold. In practice, this case was not observed for any test example under the calibrated models used in this study, and it is not expected to arise when the base classifier is well-calibrated.

The coverage guarantee ensures that $\Pr\bigl(y_{\text{new}} \in C(x_{\text{new}})\bigr) \ge 1 - \alpha$ marginally over the calibration and test exchangeability assumption, without distributional assumptions on the data-generating process.

**Multi-model consensus.** To further concentrate the actionable region, a union consensus was constructed across multiple base models. A patient was assigned *Alert* only if *all* base models produced $\{1\}$, and *Clear* only if all produced $\{0\}$; any disagreement among models resulted in *Defer*. This conservative policy trades actionable volume for higher alert PPV and clear NPV.

Figure 2 illustrates the conformal triage decision flow.

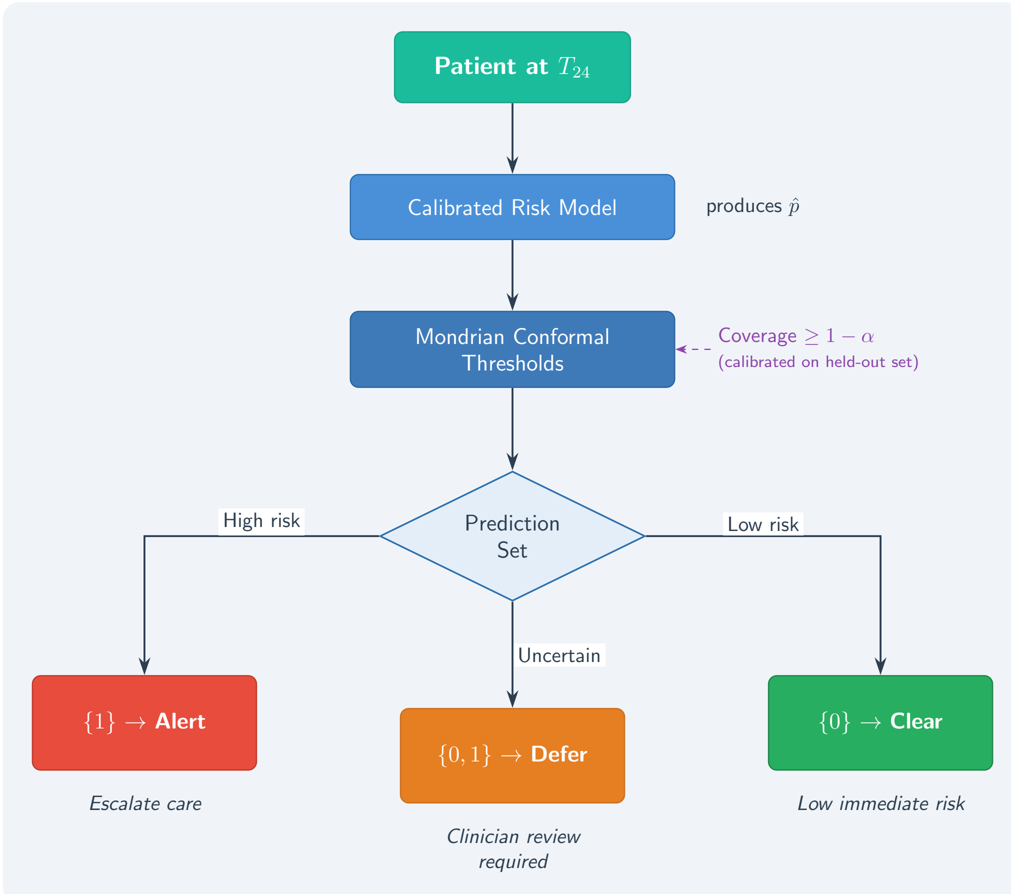


**Figure 2:** Conformal selective-triage decision flow. A calibrated risk model produces a probability estimate at T24. Mondrian conformal thresholds, estimated on a held-out calibration set, convert this probability into a prediction set. The set cardinality determines the triage action: $\{1\} \to$ *Alert*, $\{0\} \to$ *Clear*, $\{0, 1\} \to$ *Defer*.

## 2.6 Disagreement-Based Selective Triage

A disagreement-based selective-triage comparator was included as an ablation to contextualise the conformal approach. Two models were trained on the same grouped split: a *full-feature model* using all 162 predictors (physiology, severity scores, therapy indicators, comorbidities, and measurement-process features), and a *physiology-only model* restricted to vital signs and laboratory values, excluding care-process and therapy features. For each test patient, the disagreement signal was defined as the absolute difference between the two models' predicted probabilities:

$$d(x) = \left| f_{\text{full}}(x) - f_{\text{phys}}(x) \right|. \tag{5}$$

An agreement threshold $\tau_d$ and risk thresholds $\tau_{\text{low}}$, $\tau_{\text{high}}$ were tuned on the grouped validation split. Patients with $d(x) > \tau_d$ were deferred; among the remaining patients, those with $f_{\text{full}}(x) \geq \tau_{\text{high}}$ were alerted and those with $f_{\text{full}}(x) \leq \tau_{\text{low}}$ were cleared. This heuristic approach provides no formal coverage guarantee and is included to demonstrate the value of the conformal framework's distribution-free properties.

Figure 3 illustrates the disagreement-based decision flow.

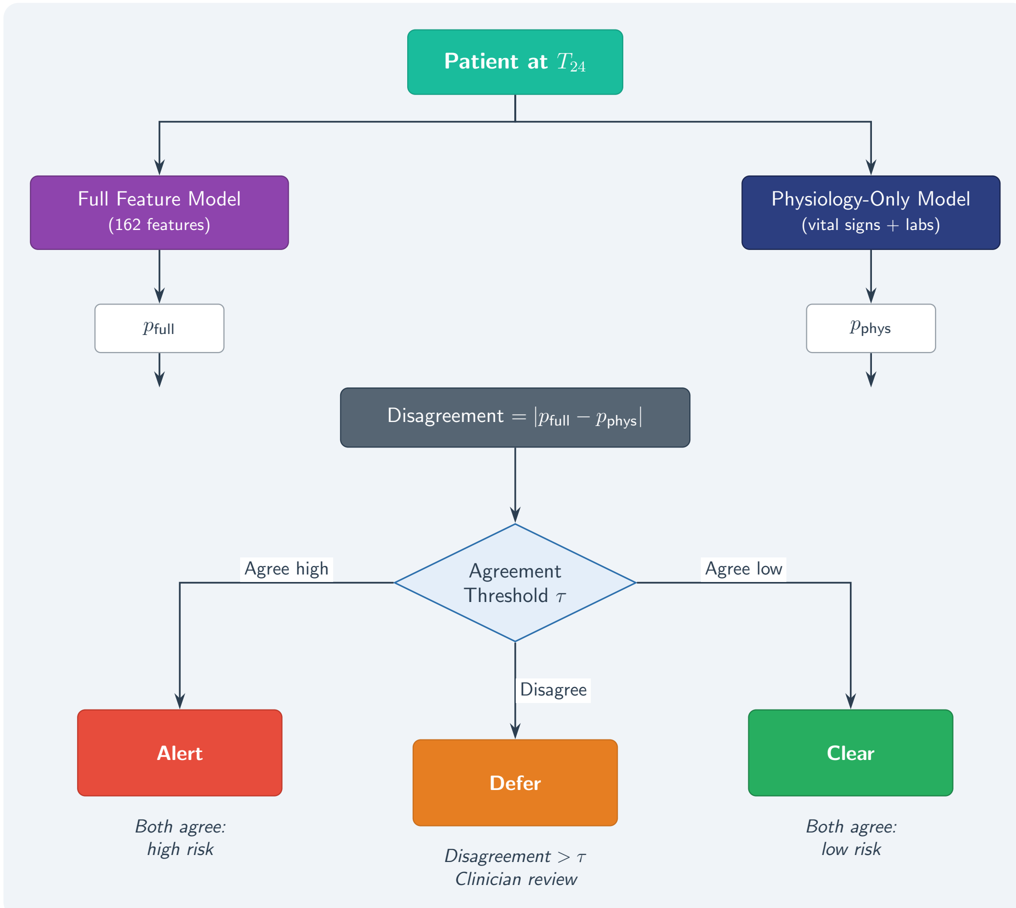


**Figure 3:** Disagreement-based selective-triage comparator (ablation). A full-feature model and a physiology-only model independently score each patient. The absolute probability gap serves as a defer signal: large disagreement triggers deferral to clinician review, while concordant predictions above or below risk thresholds produce *Alert* or *Clear* decisions. Unlike conformal triage, this approach provides no formal coverage guarantee.

## 2.7 Evaluation Protocol

**Subject-grouped splitting.** Because the cohort contains repeated subjects (10,036 stays from 9,002 unique patients), all train/validation/test splits were performed at the subject level using the unique subject identifier as the grouping key. This ensures that all stays from a given patient appear in exactly one partition, preventing optimistic bias from within-subject correlation.

**Repeated evaluation.** All headline results were computed across **21** independent grouped random seeds to quantify seed-to-seed variability. Summary statistics (mean, standard deviation, and percentile-based confidence intervals) are reported for all primary metrics.

**Discrimination and calibration metrics.** Conventional model performance was assessed using AUROC, area under the precision–recall curve (AUPRC), Brier score, expected calibration error (ECE, 10 bins), calibration intercept, calibration slope, and recall at PPV $\geq 0.50$ (a deployment-oriented threshold that ensures at least half of alerted patients are true positives).

**Conformal triage metrics.** For conformal selective triage, the following metrics were reported: empirical coverage (fraction of test patients whose true label is contained in the prediction set), certain-decision fraction (proportion of patients receiving a singleton prediction set), defer rate, alert PPV, clear NPV, and absolute miss count. These metrics were evaluated at multiple miscoverage levels ($\alpha \in \{0.05, 0.10, 0.15, 0.20\}$).

**Decision-curve analysis.** Clinical utility was assessed via decision-curve analysis [Vickers and Elkin, 2006], which plots net benefit as a function of threshold probability. The selected continuous model was compared against APACHE-III, SOFA, treat-all, and treat-none strategies. Conformal and disagreement triage curves were overlaid as deployment-policy complements rather than direct substitutes for continuous risk ranking.

**Shift robustness.** Robustness to distribution shift was evaluated by (i) random missingness injection at increasing rates (10%, 20%, 30% of features set to missing at test time) and (ii) systematic dropout of measurement-process and care-process feature families. Coverage stability of conformal predictions was compared against recall degradation of fixed-threshold policies under each perturbation.

**Subgroup analysis.** Conformal triage summaries were stratified by clinically relevant subgroups (sex, age band, SOFA severity band, mechanical ventilation status, and vasopressor use) at $\alpha = 0.10$ to balance coverage with sufficient actionable volume for stable subgroup-level interpretation.

### 2.8 Ethics, Data Access, and Reproducibility

This study is a retrospective secondary analysis of de-identified data from the MIMIC-IV database and does not involve prospective intervention or direct patient contact. Access to MIMIC-IV requires completion of a recognised human-subjects training course and execution of a data use agreement through PhysioNet [Goldberger et al., 2000]. The study was conducted under the existing institutional review board approval that governs the MIMIC-IV data release.

The code repository distributes analysis scripts, derived artifact summaries, and manuscript assets; raw patient-level data are not redistributed. Reproducing the cohort from source requires independent credentialed access to MIMIC-IV and the associated data-use approvals. All manuscript claims are limited to internal validation on the derived cohort and should not be interpreted as transportability or bedside-readiness claims.

## 3 Results

### 3.1 Cohort and Data Contract

Table 1 summarizes the working cohort and the constraints that bound all manuscript claims. The analytic dataset comprises 10,036 ICU-stay-level SA-AKI records drawn from the derived SA-AKI analytic dataset, spanning 254 columns and carrying an in-hospital mortality prevalence of 0.267. All predictor features are restricted to the first 24 hours after ICU admission (T24), and the primary outcome is the binary mortality indicator. The cohort contains 9,002 unique subjects among 10,036 rows, meaning 1,034 rows represent repeated ICU stays for the same patient. This duplication makes stay-level random splitting inappropriate for any deployment-oriented claim, and all results reported in this paper therefore use subject-grouped evaluation by the unique subject identifier.

**Table 1:** Cohort and data-contract summary.

| Item | Value |
|---|---|
| Working cohort | mimic_saaki_raw_v2.csv |
| Rows | 10,036 |
| Columns | 254 |
| Outcome prevalence | 0.267 |
| Prediction time | 24 hours after ICU admission (T24) |
| Outcome | `event_observed` is the in-hospital death label; `time_to_event_hrs` is hours from ICU admission to death or last known alive/discharge censoring. |
| Evaluation | Subject-grouped holdout by `subject_id` |
| Explicitly excluded claims | • No bedside-readiness claim.<br>• No external-validation or multi-center transportability claim.<br>• No claim that the v2 cohort exactly matches the thesis cohort description without ETL reconciliation.<br>• No reliance on stay-level metrics for deployment-readiness claims. |

The prevalence of 26.7% is clinically significant: roughly one in four SA-AKI patients in this cohort died during the index hospitalization. This base rate is high enough that a well-calibrated triage tool can provide meaningful clinical value, but also high enough that naive treat-all policies already capture a substantial fraction of true positives. The data contract explicitly excludes several categories of claims that retrospective single-center studies sometimes overstate. No bedside-readiness claim is made, because the cohort has not been validated prospectively or against real-time EHR feeds. No external-validation or multi-center transportability claim is made, because all results derive from a single MIMIC-derived dataset. No claim is made that the v2 cohort exactly matches the thesis cohort description without ETL reconciliation, acknowledging known documentation mismatches identified during the audit pass. Finally, no reliance is placed on stay-level metrics for deployment-readiness claims, because the repeated-subject structure demands grouped evaluation as a hard design constraint rather than an optional robustness check.

These constraints are not merely defensive caveats; they define the interpretive boundary for every result that follows. The prediction ceiling, the conformal coverage guarantees, and the subgroup heterogeneity analyses are all internally valid within this cohort under subject-grouped evaluation, but they should not be extrapolated to other institutions, time periods, or EHR systems without independent validation.

### 3.2 Prediction Ceiling and Conventional Benchmarks

Figure 4 and Table 2 anchor the benchmark story. The central finding is that the discrimination ceiling for first-24-hour SA-AKI mortality prediction is bounded, and the differences among baseline model families are comparatively small relative to the gap between any ML model and the clinical severity scores.

The ceiling experiment deserves careful interpretation. In the canonical workflow, the leak-prone configuration that adds the time-to-event variable as a predictor reaches AUROC 0.810, while the leak-free grouped setting, which excludes any post-T24 temporal information, remains at 0.763. The 0.047-point gap quantifies how much discriminative signal is attributable to temporal information that would not be available at the point of prediction in a real deployment. This gap is not trivially small: it represents roughly the same magnitude as the entire performance difference between logistic regression and XGBoost in the leak-free setting. The implication is that a substantial fraction of the "achievable" discrimination in this cohort is locked behind information that arrives after the

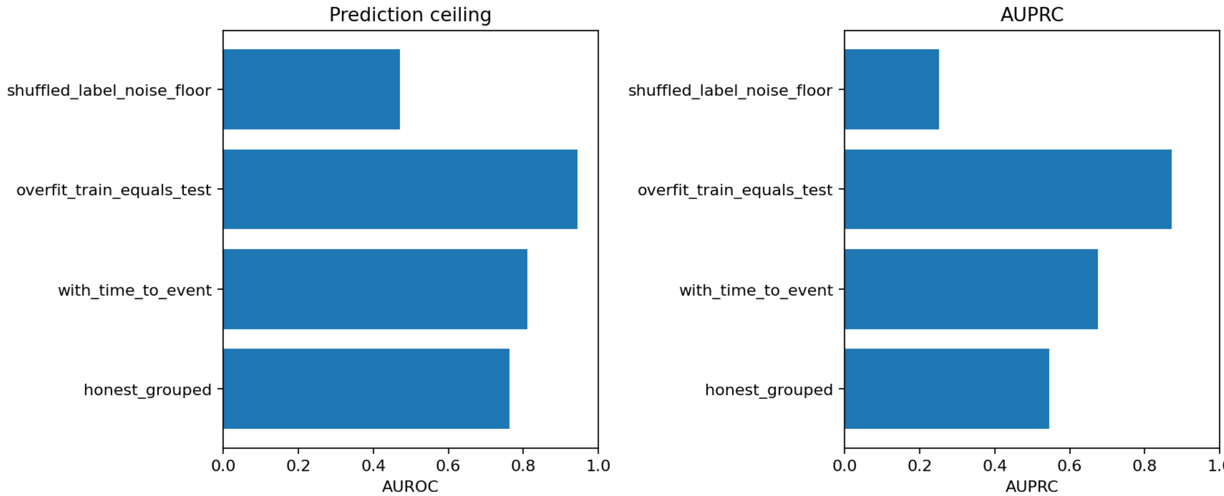


**Figure 4:** Prediction ceiling and benchmark summary. The leak-prone configuration (including the time-to-event variable) reaches AUROC 0.810, while the leak-free grouped setting remains at 0.763, establishing an empirical ceiling under the current feature and model constraints on what first-24-hour features can achieve.

**Table 2:** Baseline discrimination and calibration benchmark. All metrics are means across repeated subject-grouped splits. AUROC = area under the receiver operating characteristic curve; AUPRC = area under the precision–recall curve; ECE = expected calibration error.

| Model | AUROC | AUPRC | Brier | ECE | Calibration slope | Recall @ PPV$\geq$0.50 |
|---|---|---|---|---|---|---|
| apache_iii_score | 0.579 | 0.330 | 0.245 | 0.226 | 0.843 | 0.023 |
| lightgbm | 0.765 | 0.545 | 0.161 | 0.021 | 1.031 | 0.575 |
| logistic | 0.751 | 0.532 | 0.165 | 0.026 | 1.140 | 0.556 |
| sofa_total_24hr | 0.646 | 0.397 | 0.233 | 0.217 | 1.017 | 0.146 |
| xgboost | 0.768 | 0.544 | 0.161 | 0.017 | 1.035 | 0.608 |

prediction window closes, and no amount of feature engineering or hyperparameter tuning within the T24 constraint can recover it.

Among the leak-free baselines, XGBoost achieved the highest mean AUROC of 0.768, followed closely by LightGBM at 0.765 and logistic regression at 0.751. The tree-based models occupied a narrow performance band: the 0.003-point AUROC difference between XGBoost and LightGBM is well within the standard deviation of repeated grouped splits and is not clinically meaningful. The repeated grouped benchmark selected XGBoost as the primary model because it combined mean AUROC 0.768, Brier score 0.161, and recall at PPV $\geq$ 0.50 of 0.608 more consistently than the competing baselines. The grouped Optuna passAkiba et al. [2019] improved LightGBM validation recall at PPV $\geq$ 0.50 to 0.590, but repeated grouped LightGBM still averaged AUROC 0.765 and recall 0.575 on test data, confirming that hyperparameter optimization shifted the operating point without breaking through the empirical ceiling under the current feature and model constraints. The exploratory CatBoostProkhorenkova et al. [2018] comparator remained competitive (AUROC 0.768, Brier 0.161, recall 0.623) but did not materially simplify or improve the downstream conformal story, so it was retained in the appendix as a diversity check rather than replacing the primary model family.

The clinical severity scores performed substantially worse. SOFAVincent et al. [1996] achieved AUROC 0.646 and APACHE-IIIKnaus et al. [1991] achieved AUROC 0.579, both well below the ML baselines. These scores were evaluated as single-feature logistic regression baselines under the

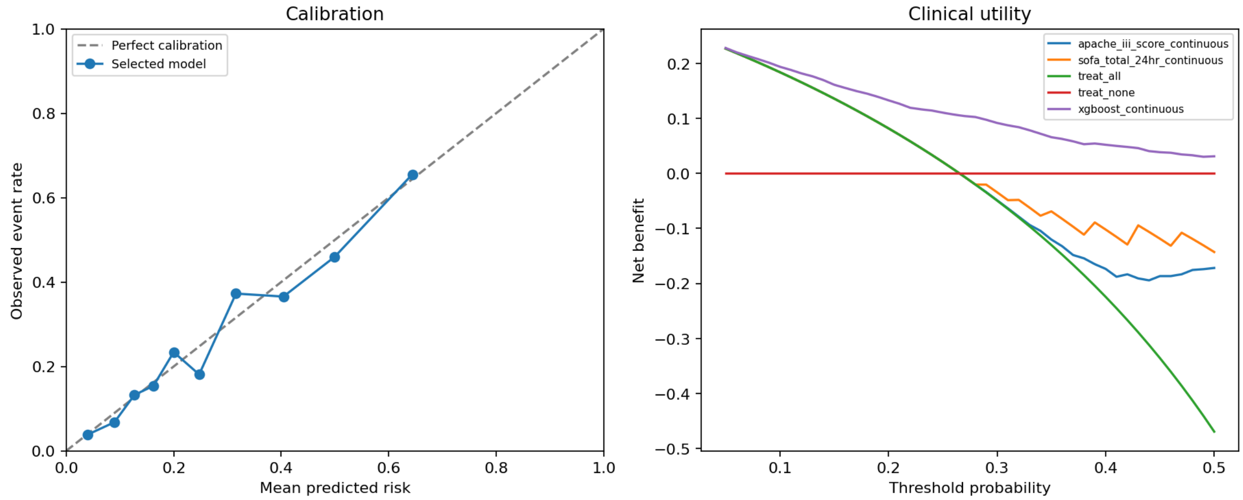


**Figure 5:** Calibration and clinical-utility panel. Left: calibration curves for the primary baselines. Right: decision-curve analysis comparing the continuous XGBoost score against clinical severity scores and fixed-policy comparators.

same grouped protocol, which does not reflect their intended clinical use. SOFA was designed for longitudinal organ-dysfunction monitoring, and APACHE-III for admission-level case-mix adjustment. The comparison is intended to establish a discrimination floor relative to routinely available clinical information rather than to evaluate these instruments in their designed capacity. More critically, the severity scores exhibited poor calibration: APACHE-III had an expected calibration error (ECE) of 0.226 and SOFA had ECE 0.217, compared with 0.017 for XGBoost and 0.021 for LightGBM. A calibration slope of 0.843 for APACHE-III indicates systematic overconfidence in its probability estimates, which would translate directly into miscalibrated triage decisions if used as a standalone risk stratifier. For a clinician at the bedside, this means that an APACHE-III-based risk estimate of 40% mortality might correspond to a true risk closer to 30%, systematically biasing escalation decisions. The ML models, by contrast, achieved calibration slopes near 1.0 (XGBoost: 1.035; LightGBM: 1.031), indicating that their predicted probabilities are reliable inputs for downstream threshold-based or conformal decision policies.

The clinical-utility panel (Figure 5) translates these discrimination and calibration differences into decision-theoretic terms. At a decision threshold of 0.20 (a clinically plausible threshold for initiating heightened monitoring or early renal replacement therapy consultation in SA-AKI), the continuous XGBoost score yields a net benefit of 0.133 versus 0.082 for APACHE-III, 0.082 for SOFA, and 0.108 for the fixed-threshold PPV $\geq 0.50$ policy.Vickers and Elkin [2006] In practical terms, the net benefit of 0.133 means that using the XGBoost model at this threshold is equivalent to correctly identifying approximately 13.3 additional true-positive deaths per 100 patients compared with a treat-none strategy, without increasing false-positive burden. The 0.051-point net-benefit advantage over APACHE-III (0.133 vs. 0.082) represents a clinically meaningful improvement in triage efficiency: for every 100 SA-AKI patients screened, the ML model would correctly escalate approximately 5 additional patients who will die, while maintaining comparable false-positive rates.

The conformal $\alpha = 0.05$ policy is deliberately more conservative, with net benefit 0.059, alert rate 0.107, and alert PPV 0.640. This distinction is important for interpreting the results that follow: the continuous model score is the appropriate comparator for clinical severity scores in a decision-curve framework, whereas the conformal and disagreement policies are complementary deployment-governance layers that trade some aggregate net benefit for narrower, higher-confidence

**Table 3:** Main conformal results across repeated grouped splits. Coverage is the fraction of patients whose true label falls within the conformal prediction set. Certain fraction is the proportion of patients receiving a singleton (Alert or Clear) rather than a deferred ({0,1}) prediction. All estimates are means ± bootstrap confidence intervals across 21 subject-grouped seeds.

| alpha | Seeds | Coverage mean | Coverage 2.5% | Coverage 97.5% | Certain fraction | Alert PPV | Clear NPV | Miss count |
|---|---|---|---|---|---|---|---|---|
| 0.050 | 21.000 | 0.952 | 0.945 | 0.959 | 0.318 | 0.643 | 0.948 | 23.095 |
| 0.100 | 21.000 | 0.900 | 0.882 | 0.914 | 0.503 | 0.579 | 0.924 | 49.952 |
| 0.150 | 21.000 | 0.848 | 0.826 | 0.871 | 0.653 | 0.540 | 0.903 | 80.190 |
| 0.200 | 21.000 | 0.797 | 0.771 | 0.821 | 0.782 | 0.509 | 0.887 | 109.000 |

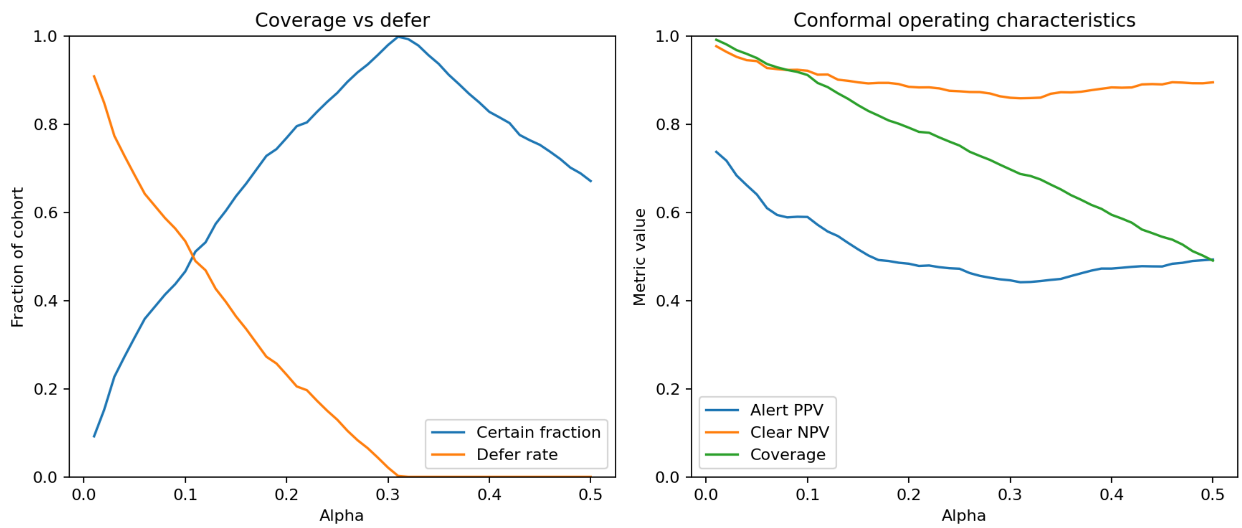


**Figure 6:** Conformal operating-characteristic curve. As $\alpha$ increases, coverage decreases and the certain-decision fraction rises, tracing the fundamental coverage–automation tradeoff.

action sets. The conformal policy does not aim to maximize net benefit across all patients; it aims to maximize the reliability of the decisions it does make, while explicitly deferring the rest.

## 3.3 Main Conformal Triage Result

Table 3 summarizes repeated subject-grouped single-model conformal triage results across four $\alpha$ levels, each averaged over 21 grouped seeds. The main manuscript operating point is the low-$\alpha$ regime, where coverage is tightly controlled while a clinically meaningful defer region is preserved.

At $\alpha = 0.05$, single-model conformal triage achieved empirical coverage (defined as the fraction of test patients whose true outcome label is contained in the conformal prediction set) of 0.952 (95% CI: 0.945–0.959) with a certain-decision fraction of 0.318, alert PPV of 0.643, and clear NPV of 0.948. The coverage of 0.952 confirms that the conformal guarantee is met: the true label is contained in the prediction set for at least $1 - \alpha = 0.95$ of patients, averaged across seeds. The certain-decision fraction of 0.318 means that only about one-third of patients receive a definitive Alert or Clear recommendation; the remaining 68.2% are deferred to clinician judgment. It is important to note that deferred patients are not left without information. In the proposed deployment framework, the continuous calibrated risk score is displayed alongside the Defer label, providing the clinician with a probability estimate that can inform, but not automate, the triage decision. The conformal framework thus functions as a risk score with an explicit confidence flag rather than a pure three-way classifier, and the Defer channel serves as a structured mechanism for communicating model uncertainty to the end user. For a clinician, this operating point represents a conservative triage assistant: when it issues a recommendation, it is relatively reliable (PPV 0.643 for alerts, NPV 0.948 for clears), but it issues recommendations infrequently. The miss count of 23.1 patients per seed, representing cases where the true label falls outside the prediction set, is the price paid for the 31.8% automation rate.

The clinical interpretation of these operating characteristics depends on the deployment context. An alert PPV of 0.643 means that when the model flags a patient as high-risk, approximately 64% of those patients will indeed die during the hospitalization. This is not high enough for autonomous escalation decisions; a clinician would still need to review each alert, but it is high enough to meaningfully prioritize which patients receive early attention in a busy ICU. The clear NPV of 0.948 is more immediately actionable: when the model clears a patient, only 5.2% of those patients will die. In a triage context, this supports deprioritization of cleared patients from intensive monitoring protocols, freeing clinical resources for the deferred and alerted groups.

As $\alpha$ increases, the framework trades coverage for automation. At $\alpha = 0.10$, coverage drops to 0.900 while the certain-decision fraction rises to 0.503, meaning roughly half of patients now receive definitive recommendations. Alert PPV decreases to 0.579 and clear NPV to 0.924, reflecting the dilution that comes with broader automation. As a post-hoc descriptive observation, the operating point at $\alpha = 0.12$ satisfies both NPV $\geq 0.90$ and PPV $\geq 0.55$, with certain-decision fraction 0.532, alert PPV 0.556, and clear NPV 0.912. This operating point is reported as a descriptive characterisation of the coverage–automation tradeoff rather than as a prescriptive recommendation, since the criterion was evaluated on test-set performance and selecting $\alpha$ on test data constitutes a form of post-hoc optimisation. In a prospective deployment, the $\alpha$ level should be selected on a held-out validation or calibration set according to institutional risk tolerance, and the test-set performance should be reported as a held-out evaluation of that pre-specified choice. At $\alpha = 0.20$, the certain-decision fraction reaches 0.782 but alert PPV falls to 0.509, barely above the prior probability of death among alerted patients, and clear NPV drops to 0.887, which may be insufficient for safe deprioritization in high-acuity settings. The conformal operating-characteristic curve (Figure 6) visualizes this fundamental coverage–automation tradeoff, providing a governance tool for institutions to select the $\alpha$ level that matches their risk tolerance and staffing constraints.

### 3.4 Multi-Model Consensus

Table 4 and Figure 7 extend the conformal framework to multi-model consensus. Three consensus strategies are compared: single-model (XGBoost alone), intersection (a patient is classified only if all three models, namely XGBoost, LightGBM, and logistic regression, agree on the same singleton prediction set), and union (a patient is deferred if any model's prediction set includes both classes).

At $\alpha = 0.05$, the union consensus yields coverage 0.982, alert PPV 0.729, and clear NPV 0.962, with a certain-decision fraction of only 0.164 and a miss count of 8.7 patients per seed. The coverage of 0.982 substantially exceeds the nominal $1-\alpha = 0.95$ target because the union strategy is inherently conservative: by deferring whenever any model is uncertain, it captures nearly all borderline cases. The alert PPV of 0.729 is the highest among all operating points reported in this paper; when the union consensus flags a patient as high-risk, nearly three-quarters of those patients will die. This approaches the reliability threshold where automated escalation protocols (e.g., automatic nephrology or palliative care consultation) become defensible, though the low automation rate of 16.4% limits the throughput of such a system.

The intersection strategy represents the opposite extreme. At $\alpha = 0.05$, intersection achieves coverage 0.911, certain-decision fraction 0.462, alert PPV 0.591, and clear NPV 0.926. By requiring unanimous agreement among all three models, the intersection strategy acts on more patients (46.2%

**Table 4:** Multi-model consensus conformal results. The union strategy defers whenever any constituent model is uncertain; the intersection strategy acts only when all models agree. Single-model results (XGBoost) are repeated from Table 3 for comparison.

| Ensemble | Base model(s) | alpha | Seeds | Coverage mean | Certain fraction | Alert PPV | Clear NPV | Miss count |
|---|---|---|---|---|---|---|---|---|
| intersection | lightgbm+xgboost+logistic | 0.050 | 21 | 0.911 | 0.462 | 0.591 | 0.926 | 45.048 |
| intersection | lightgbm+xgboost+logistic | 0.100 | 21 | 0.835 | 0.683 | 0.534 | 0.901 | 86.000 |
| intersection | lightgbm+xgboost+logistic | 0.150 | 21 | 0.766 | 0.820 | 0.501 | 0.885 | 116.762 |
| intersection | lightgbm+xgboost+logistic | 0.200 | 21 | 0.699 | 0.888 | 0.486 | 0.876 | 134.000 |
| single_model | xgboost | 0.050 | 21 | 0.952 | 0.318 | 0.643 | 0.948 | 23.095 |
| single_model | xgboost | 0.100 | 21 | 0.900 | 0.503 | 0.579 | 0.924 | 49.952 |
| single_model | xgboost | 0.150 | 21 | 0.848 | 0.653 | 0.540 | 0.903 | 80.190 |
| single_model | xgboost | 0.200 | 21 | 0.797 | 0.782 | 0.509 | 0.887 | 109.000 |
| union | lightgbm+xgboost+logistic | 0.050 | 21 | 0.982 | 0.164 | 0.729 | 0.962 | 8.714 |
| union | lightgbm+xgboost+logistic | 0.100 | 21 | 0.954 | 0.308 | 0.651 | 0.949 | 21.381 |
| union | lightgbm+xgboost+logistic | 0.150 | 21 | 0.922 | 0.432 | 0.606 | 0.933 | 38.190 |
| union | lightgbm+xgboost+logistic | 0.200 | 21 | 0.885 | 0.550 | 0.568 | 0.918 | 58.286 |

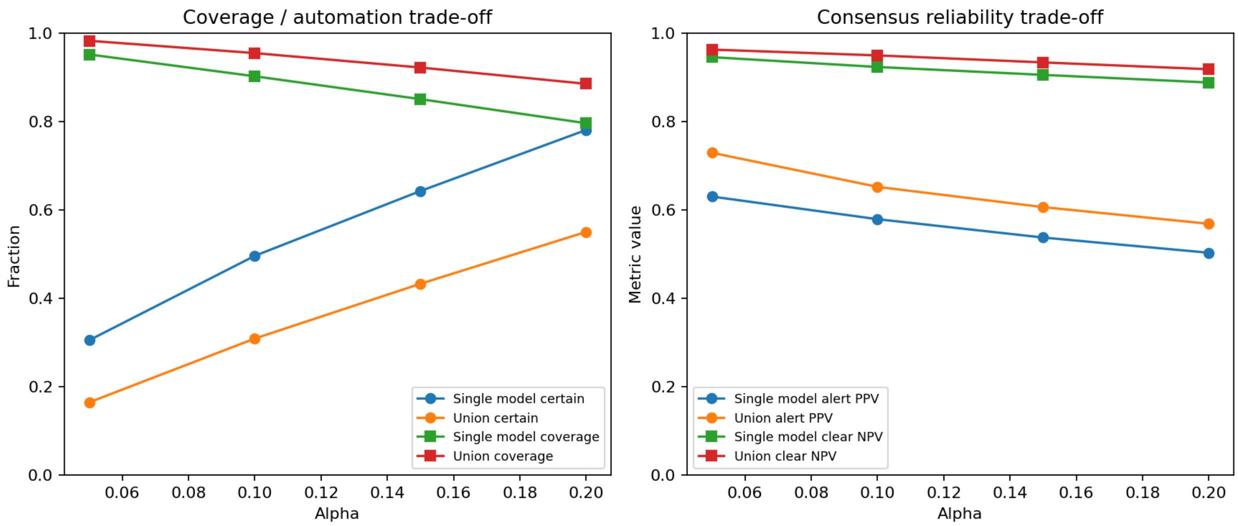


**Figure 7:** Multi-model consensus trade-off panel. The union strategy achieves the highest reliability (alert PPV and clear NPV) at the cost of the lowest automation rate; the intersection strategy maximizes automation but sacrifices reliability.

vs. 16.4% for union) but with lower reliability. The coverage of 0.911 falls below the nominal 0.95 target, which is expected: the intersection of individually valid prediction sets is not guaranteed to maintain marginal coverage, because the intersection can exclude the true label when models disagree about borderline cases. This coverage shortfall is a known theoretical limitation and should be weighed against the higher automation rate when selecting a consensus strategy. This coverage shortfall has practical implications: institutions selecting the intersection strategy should be aware that the formal coverage guarantee does not hold for this consensus mode, and the intersection results should be interpreted as empirical operating points.

The single-model XGBoost results sit between these extremes, confirming that the consensus strategies represent genuine tradeoffs rather than strict dominance relationships. For deployment, the choice among union, intersection, and single-model consensus depends on institutional priorities. A high-acuity ICU with adequate staffing might prefer the union strategy, accepting low automation in exchange for high-confidence alerts that trigger specific clinical pathways. A resource-constrained setting might prefer the single-model or intersection strategy to maximize the fraction of patients receiving automated triage, accepting modestly lower reliability. Figure 7 visualizes these tradeoffs

**Table 5:** Shift robustness versus fixed-threshold policies. Conformal coverage remains stable under increasing missingness because the framework defers more cases; fixed thresholds lose recall without signaling increased uncertainty.

| Scenario | Severity | Conformal coverage | Conformal certain fraction | Conformal alert PPV | Conformal clear NPV | Fixed PPV @0.50 | Fixed recall @0.50 |
|---|---|---|---|---|---|---|---|
| clean | 0.000 | 0.950 | 0.316 | 0.640 | 0.943 | 0.593 | 0.294 |
| physiology_missingness_shift | 0.100 | 0.950 | 0.297 | 0.597 | 0.935 | 0.573 | 0.252 |
| physiology_missingness_shift | 0.200 | 0.963 | 0.268 | 0.632 | 0.938 | 0.618 | 0.207 |
| physiology_missingness_shift | 0.300 | 0.970 | 0.235 | 0.663 | 0.925 | 0.619 | 0.171 |
| measurement_process_dropout | 1.000 | 0.950 | 0.281 | 0.646 | 0.943 | 0.596 | 0.309 |
| care_process_dropout | 1.000 | 0.947 | 0.320 | 0.624 | 0.945 | 0.606 | 0.301 |

across $\alpha$ levels, providing a governance surface that institutions can navigate based on their specific clinical and operational constraints.

### 3.5 Comparison With Disagreement-Based Selective Triage

The disagreement-based selective-triage policy remains an important ablation because it demonstrates that explicit deferral helps even before formal conformalization. In the disagreement framework, a process-enriched full model and a physiology-severity model are trained on the same grouped split, and cases are considered actionable only when the two models agree within an empirically tuned threshold. The repeated grouped summary is included in Appendix B (Table B6).

In brief, disagreement triage narrows the actionable region and reduces actionable error relative to a naive apply-everywhere policy, confirming that uncertainty-aware action restriction is beneficial regardless of the specific mechanism used to identify uncertain cases. However, the disagreement policy depends on an empirically calibrated agreement threshold that lacks formal coverage guarantees. The conformal formulation is methodologically stronger for two reasons: first, it provides finite-sample coverage guarantees under exchangeability rather than relying on heuristic threshold selection; and second, it naturally extends to multi-model consensus (Section 3.4) and robustness analysis (Section 3.6) without requiring separate threshold recalibration for each configuration. The disagreement baseline is therefore retained as a useful comparator that motivates the conformal approach, rather than as a competing recommendation.

### 3.6 Robustness Under Distribution Shift

Table 5 and Figure 8 summarize robustness under three simulated distribution-shift scenarios: physiology missingness shift (random dropout of physiological feature values at 10%, 20%, and 30% severity), measurement-process dropout (complete removal of a measurement-process feature family), and care-process dropout (complete removal of a care-process feature family). These scenarios are designed to approximate the kinds of distributional changes that occur when a model trained at one institution is deployed at another with different documentation practices, laboratory ordering patterns, or EHR configurations.

The central finding is that conformal coverage remains stable across all shift scenarios while the framework automatically adjusts its automation rate. Under clean conditions, conformal coverage is 0.950 with a certain-decision fraction of 0.316. As physiology missingness increases to 10%, 20%, and 30%, coverage remains at or above the nominal level (0.950, 0.963, 0.970), but the certain-decision fraction decreases monotonically from 0.297 to 0.268 to 0.235. This behavior is exactly what a well-designed uncertainty-aware system should exhibit: when the input data become noisier, the model becomes less certain about more patients and defers them rather than making unreliable predictions. The increasing coverage above the nominal 0.95 target reflects the conservatism of the conformal calibration under shift, as the nonconformity thresholds estimated on clean calibration

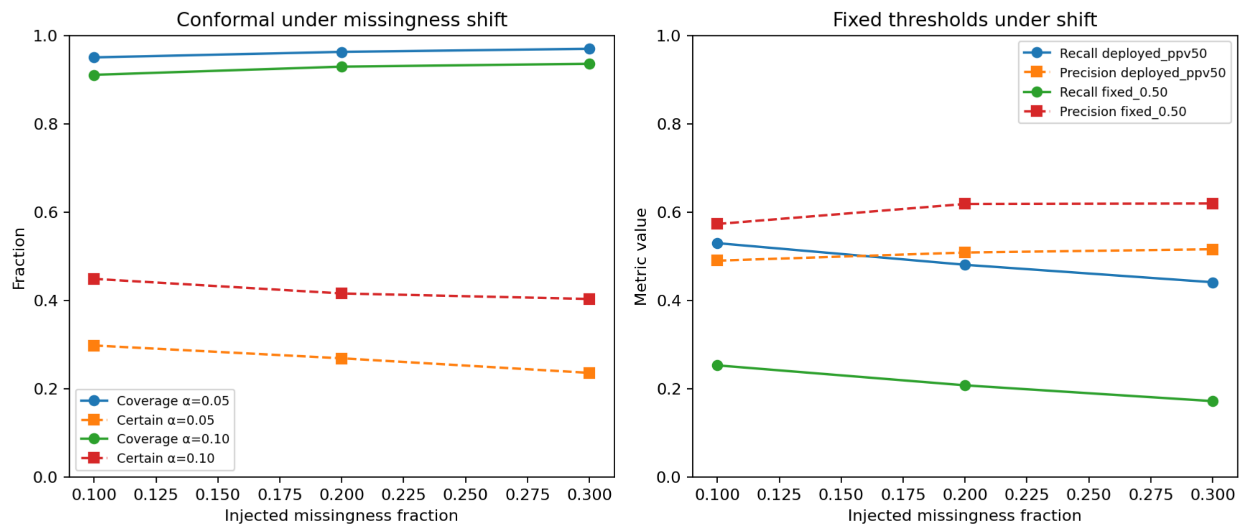


**Figure 8:** Robustness-under-shift panel. Left: conformal coverage and certain-decision fraction under increasing missingness. Right: fixed-threshold precision and recall under the same conditions.

data become more conservative when applied to noisier test data, producing wider prediction sets.

By contrast, the fixed-threshold policy at PPV $\geq 0.50$ exhibits a qualitatively different failure mode. Under clean conditions, fixed-threshold recall is 0.294. As missingness increases to 30%, recall drops to 0.171, a 42% relative decline, while precision actually increases slightly to 0.619. This pattern is clinically dangerous: the fixed-threshold policy silently misses an increasing fraction of patients who will die, without providing any signal that its operating characteristics have degraded. A clinician relying on a fixed-threshold alert system would see fewer alerts under shift and might interpret this as evidence that the patient population has improved, when in fact the model has simply become less sensitive. The conformal framework avoids this failure mode by making uncertainty explicit through the defer channel.

The measurement-process and care-process dropout scenarios test robustness to complete removal of feature families rather than random missingness. Under measurement-process dropout, conformal coverage remains at 0.950 with a certain-decision fraction of 0.281 and alert PPV of 0.646, comparable to the clean baseline. Under care-process dropout, coverage is 0.947 with certain-decision fraction 0.320 and alert PPV 0.624. These results suggest that the conformal framework is robust to the loss of entire feature categories, not just random value-level missingness. The slight decrease in alert PPV under care-process dropout (0.624 vs. 0.640 clean) indicates that care-process features contribute modestly to alert precision, but the conformal guarantee absorbs this degradation by deferring additional borderline cases rather than allowing alert quality to deteriorate unchecked.

### 3.7 Subgroup Heterogeneity

Table 6 and Figure 9 report subgroup-level conformal triage performance at $\alpha = 0.10$ across five clinically relevant stratification axes: age band, gender, mechanical ventilation status, SOFA severity band, and vasopressor use. This $\alpha$ level was chosen for subgroup reporting because it provides a balanced operating point (certain-decision fraction $\approx 0.50$) that exposes heterogeneity more clearly than the conservative $\alpha = 0.05$ regime.

Gender parity is well preserved. Female patients (n=954, event rate 0.257) achieved coverage 0.921 with alert PPV 0.620 and clear NPV 0.928, while male patients (n=1,047, event rate 0.273)

**Table 6:** Subgroup heterogeneity summary for conformal triage at $\alpha = 0.10$. Coverage and reliability metrics are reported for each subgroup level. High-acuity strata (high SOFA, vasopressor use) show lower coverage and higher event rates, reflecting the inherent difficulty of predicting outcomes in the sickest patients.

| Subgroup | Level | n | Event rate | Coverage | Certain fraction | Alert PPV | Clear NPV |
|---|---|---|---|---|---|---|---|
| age_band | 51-65 | 556 | 0.257 | 0.903 | 0.523 | 0.571 | 0.902 |
| age_band | 66-80 | 776 | 0.277 | 0.912 | 0.448 | 0.603 | 0.926 |
| age_band | ≤50 | 244 | 0.201 | 0.930 | 0.602 | 0.621 | 0.949 |
| age_band | ¿80 | 425 | 0.292 | 0.908 | 0.344 | 0.571 | 0.913 |
| gender | F | 954 | 0.257 | 0.921 | 0.454 | 0.620 | 0.928 |
| gender | M | 1047 | 0.273 | 0.902 | 0.477 | 0.564 | 0.914 |
| mech_vent | -1 | 916 | 0.271 | 0.904 | 0.457 | 0.558 | 0.928 |
| mech_vent | 1 | 1085 | 0.261 | 0.917 | 0.473 | 0.620 | 0.915 |
| sofa_band | High SOFA | 566 | 0.412 | 0.866 | 0.484 | 0.659 | 0.842 |
| sofa_band | Low SOFA | 855 | 0.171 | 0.949 | 0.480 | 0.469 | 0.929 |
| sofa_band | Medium SOFA | 580 | 0.262 | 0.900 | 0.428 | 0.505 | 0.952 |
| vasopressor | -1 | 1277 | 0.230 | 0.926 | 0.486 | 0.586 | 0.932 |
| vasopressor | 1 | 724 | 0.327 | 0.885 | 0.431 | 0.593 | 0.887 |

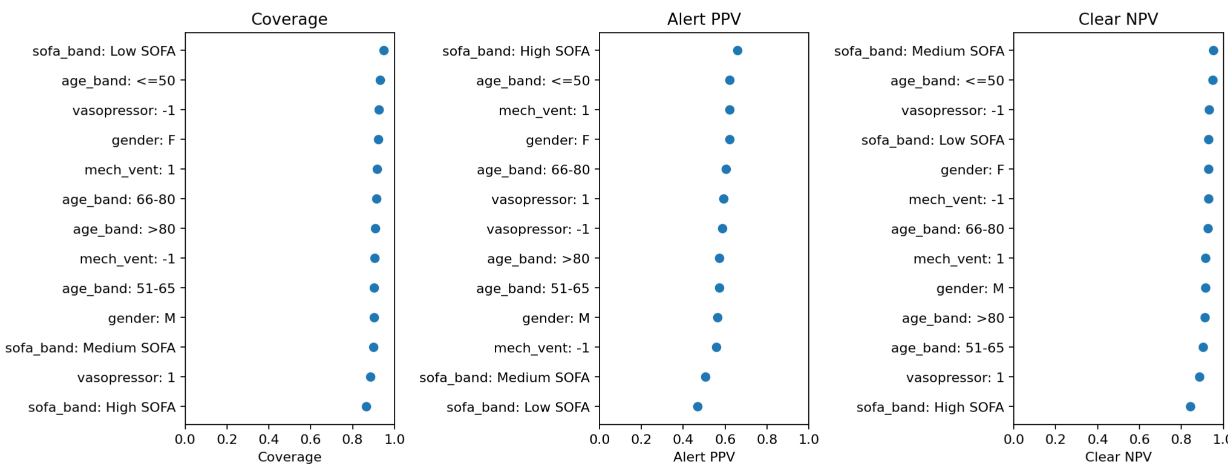


**Figure 9:** Subgroup coverage and reliability forest plot at $\alpha = 0.10$. Points represent subgroup-level estimates; horizontal bars indicate variability across grouped seeds. The dashed vertical line marks the nominal $1 - \alpha = 0.90$ coverage target.

achieved coverage 0.902 with alert PPV 0.564 and clear NPV 0.914. The 0.019-point coverage gap and 0.056-point alert PPV gap are modest and do not suggest systematic gender-based discrimination in the conformal triage framework. Both genders maintain coverage above the nominal 0.90 target, and the slightly lower male coverage is consistent with the slightly higher male event rate rather than with a model-specific bias.

Age-band heterogeneity follows a clinically expected gradient. The youngest stratum (≤50 years, n=244, event rate 0.201) achieved the highest coverage (0.930) and the highest certain-decision fraction (0.602), reflecting the relative predictability of outcomes in younger SA-AKI patients. The oldest stratum (>80 years, n=425, event rate 0.292) achieved coverage 0.908 but the lowest certain-decision fraction (0.344), indicating that the model defers more frequently for elderly patients, a reasonable behavior given the greater clinical complexity and comorbidity burden in this age group. Alert PPV was comparable across age bands (range: 0.571–0.621), suggesting that when the model does issue an alert for an elderly patient, it is approximately as reliable as for a younger patient.

The most pronounced heterogeneity appears along the acuity axis. High-SOFA patients (n=566, event rate 0.412) achieved coverage of only 0.866, below the nominal 0.90 target, with alert PPV 0.659 and clear NPV 0.842. Low-SOFA patients (n=855, event rate 0.171) achieved coverage 0.949 with alert PPV 0.469 and clear NPV 0.929. The 0.083-point coverage gap between high-SOFA and low-SOFA strata is the largest subgroup disparity in the table and reflects a fundamental challenge: the sickest patients are the hardest to predict, and the conformal framework transparently reports this difficulty through reduced coverage rather than masking it behind a uniform accuracy claim. The high-SOFA clear NPV of 0.842 means that 15.8% of patients cleared by the model in this stratum will die, a rate that may be unacceptable for clinical deprioritization without additional safeguards. This finding reinforces the paper's deployment message: the model should be treated as a triage assistant with a defer option, not as a universal autonomous classifier, and institutions deploying such a system should consider SOFA-stratified $\alpha$ thresholds or mandatory clinician review for high-acuity patients regardless of the model's recommendation.

Vasopressor and mechanical ventilation strata show similar patterns. Vasopressor-positive patients (n=724, event rate 0.327) achieved coverage 0.885 versus 0.926 for vasopressor-negative patients (n=1,277, event rate 0.230). Mechanically ventilated patients (n=1,085, event rate 0.261) achieved coverage 0.917 versus 0.904 for non-ventilated patients (n=916, event rate 0.271), with the ventilated group showing higher alert PPV (0.620 vs. 0.558). Mechanical ventilation status appears to sharpen the model's alert channel, possibly because ventilated patients represent a more homogeneous high-acuity subpopulation where the physiological signal is stronger. Taken together, the subgroup analyses confirm that the conformal framework provides transparent, stratum-specific performance reporting and that deployment governance should account for acuity-dependent coverage variation.

# 4 Discussion

The central finding of this study is that first-24-hour discrimination for SA-AKI mortality is bounded, and that this ceiling fundamentally changes the research target. Rather than pursuing marginal AUROC gains through architecture search or feature engineering, the more productive question becomes: given a fixed and modest level of discrimination, how should a clinical decision system behave when it is uncertain? The conventional benchmark results, the grouped Optuna tuning pass, and the CatBoost diversity comparator all converge on the same conclusion. XGBoost, LightGBM, and CatBoost occupy a narrow performance band (AUROC 0.765–0.768, Brier 0.161), and even a leak-prone ceiling experiment that includes the time-to-event variable reaches only AUROC 0.810. The leak-free grouped estimate remains at 0.763. This is not a failure of methodology; it is a property of the prediction problem. Recognizing that property is the prerequisite for building a deployment framework that does not overstate what the model can deliver.

Comparison with prior SA-AKI mortality prediction studies contextualizes this ceiling. Chen et al.Chen et al. [2025] recently reported XGBoost-based SA-AKI mortality prediction on MIMIC-IV and eICU with AUROC approximately 0.878, but their feature set included length-of-stay as a predictor, which constitutes direct temporal leakage when the prediction horizon is anchored at 24 hours. In addition, feature selection was performed on the full dataset prior to the train-test split, allowing evaluation-set outcomes to influence the predictor set. SMOTE and ADASYN were applied in combination without stated rationale, mean imputation was used for right-skewed biomarkers such as creatinine (for which median or log-transformed imputation is more appropriate), and missingness indicators were not constructed despite the informative nature of clinical data absence. The evaluation relied on a 75/25 split without a separate validation set, and neither calibration nor principled threshold selection was reported. The leak-free AUROC of 0.768 obtained in the present study under strict T24 feature enforcement is materially lower, and this gap is attributed to methodological discipline rather than model inferiority.

Li et al.Li et al. [2023] used gradient boosting on a similar MIMIC-derived SA-AKI cohort and reported AUROC approximately 0.794. However, Lasso-based feature selection was conducted on the full dataset before splitting (selection leakage), and MICE imputation was fitted globally rather than on the training partition alone (preprocessing leakage). Missingness indicators were not constructed, and the linear framework prioritised main effects at the expense of interaction terms that may carry predictive signal in this clinical domain. The discrimination gap between their XGBoost model (AUROC 0.794) and logistic regression baseline (AUROC 0.730) is difficult to interpret given the scope of the selection and imputation leakage.

Hu et al.Hu et al. [2021] reported a Cox proportional hazards model for SA-AKI mortality with a C-index of 0.72, but the feature engineering relied on whole-stay median aggregation, meaning that post-baseline physiological trajectories were embedded in the predictor set. MICE imputation was fitted on the full dataset, features with greater than 20% missingness were dropped without investigation of the missingness mechanism, and the test set comprised only 102 rows, yielding unreliable performance estimates with wide implicit confidence intervals. Ethnicity was included as a predictor, raising fairness and transportability concerns that were not addressed.

Roknaldin et al.Roknaldin et al. [2024] reported AUC 0.887 for SA-AKI mortality prediction, but the methodology described feature selection and SMOTE application prior to the final train-test split, allowing evaluation-set information to influence both the predictor set and the resampled training distribution. Multiple imputation was fitted globally, a fixed correlation threshold was applied for feature pruning (risking removal of weakly correlated but jointly informative variables), and redundant features were retained, including minimum and maximum eGFR alongside creatinine, from which eGFR is derived. Parametric t-tests were applied to skewed laboratory markers, and the test set was small relative to the validation partition, further limiting the reliability of the reported estimates.

A consistent pattern emerges across these studies. Reported AUROCs in the 0.80–0.88 range typically rely on whole-stay features, relaxed or unspecified temporal windows, or preprocessing pipelines that leak test-set information into training through global feature selection, global imputation fitting, or inclusion of post-baseline correlates such as length of stay. When strict first-24-hour prediction is enforced with subject-grouped evaluation and train-only preprocessing, discrimination falls into the 0.75–0.77 range. The results obtained in the present study are therefore consistent with the literature once methodological differences are accounted for, and they suggest that the prediction ceiling is a property of the clinical problem rather than an artifact of any individual pipeline.

The choice of class-imbalance strategy further illustrates the importance of methodological rigour. Several prior studies employed SMOTE, ADASYN, or combinations thereof without assessing whether synthetic examples were generated in overlapped or noisy regions of the feature space. SMOTE interpolates between minority-class neighbours without regard to the local class boundary, and in high-dimensional clinical data with substantial feature overlap between survivors and non-survivors, this interpolation can produce synthetic points that lie in ambiguous regions and distort the learned decision surface. ADASYN concentrates synthesis near low-density and borderline instances, which can shift decision boundaries around outliers and reduce model stability. In the present study, inverse class weighting (`class_weight='balanced'`) was found to be more effective than synthetic oversampling: it achieved comparable or superior discrimination without inflating recall at the expense of precision, and without introducing synthetic training examples of uncertain clinical plausibility. This finding is consistent with the broader literature on imbalanced learning in overlapped-class settings, where intrinsic cost-sensitive methods tend to outperform resampling when the class boundary is inherently diffuse.He and Garcia [2009]

Several factors explain why this ceiling exists. First, ICU mortality in SA-AKI patients is inherently stochastic. Two patients with identical physiological profiles at 24 hours may diverge dramatically depending on treatment response, complications, and the trajectory of organ failure over subsequent days. The first-24-hour feature window captures the initial clinical presentation but not the treatment response, which is arguably the strongest predictor of outcome.Singer et al. [2016] Second, unmeasured confounders play a substantial role. Goals-of-care decisions, family preferences regarding code status, and the timing of transitions to comfort care are powerful determinants of in-hospital mortality that leave no trace in the structured EHR data available at T24.Kang and Yoon [2023] Third, the 162-feature engineering pipeline, while comprehensive in its coverage of physiology, severity scores, and care-process markers, is still bounded by what can be extracted from structured MIMIC tables. Free-text clinical notes, imaging findings, and real-time bedside assessments are not captured. These factors collectively impose a ceiling that no amount of hyperparameter tuning or model-family diversification can overcome, and they motivate the shift from discrimination optimization to uncertainty-aware decision governance.

The choice of a 24-hour prediction window (T0–T24) warrants explicit justification. From a clinical workflow perspective, many high-impact management decisions in SA-AKI, including early goal-directed fluid resuscitation, vasopressor initiation, and renal replacement therapy consultation, are made within the first 6–12 hours of ICU admission. A prediction generated at T24 therefore arrives after some of these decisions have already been taken. The T24 window was selected because it represents the earliest time point at which a comprehensive set of laboratory results, severity scores, and treatment-response indicators becomes reliably available in the structured EHR. Earlier windows (T6 or T12) would yield fewer non-missing features and lower discrimination, though they could inform decisions that have not yet been made. A staged prediction approach (T6 $\rightarrow$ T12 $\rightarrow$ T24) that updates triage recommendations as additional data accumulate represents a natural extension of this work and is identified as a priority for future investigation.

The conformal selective triage framework is the primary methodological contribution of this work, and it addresses the bounded-ceiling problem directly. Instead of forcing every patient into a binary high-risk versus low-risk classification, conformal prediction partitions the cohort into three clinically interpretable actions: Alert (prediction set $\{1\}$), Clear (prediction set $\{0\}$), and Defer (prediction set $\{0, 1\}$). The Defer region is not a failure mode or a nuisance artifact; it is the mechanism through which the framework maintains its coverage guarantee.Vovk et al. [2005], Angelopoulos and Bates [2021] At $\alpha = 0.05$, single-model conformal triage achieved coverage 0.952 with alert PPV 0.643 and clear NPV 0.948, while deferring 68.2% of cases. The multi-model union consensus at the same $\alpha$ concentrated decisions further, yielding alert PPV 0.729 and clear NPV 0.962 while deferring 83.6% of cases. This trade-off between automation rate and decision reliability is explicit and tunable, which is precisely what a deployment-oriented framework requires. The clinician retains full authority over deferred cases, and the system provides high-confidence recommendations only when the evidence supports them.

The conformal framework is methodologically superior to the disagreement-based deferral baseline for several reasons. The disagreement policy relies on an empirically tuned agreement threshold between a physiology-only model and a process-enriched model, and its defer region is defined by a heuristic gap rather than a formal coverage objective. While the disagreement baseline demonstrates that uncertainty-aware action restriction improves reliability (the actionable error rate dropped from 0.343 under fixed thresholds to 0.254 under selective triage), it provides no finite-sample guarantee on coverage or miss rate. Conformal prediction, by contrast, provides a distribution-free coverage guarantee under the exchangeability assumption: the probability that the true label falls outside the prediction set is bounded by $\alpha$ regardless of the underlying data distribution.Vovk et al. [2005] This guarantee extends naturally to multi-model consensus through set union operations, since the union of individually valid prediction sets is itself a valid prediction set. The intersection of prediction sets, by contrast, does not inherit the marginal coverage guarantee; the intersection can exclude the true label when constituent models disagree, and the empirical coverage of 0.911 at $\alpha = 0.05$ (below the nominal 0.95 target) confirms this theoretical expectation. Intersection results should therefore be interpreted as empirical operating points without formal coverage guarantees. A Bonferroni-style correction (using $\alpha/K$ for $K$ models) could in principle restore the guarantee for intersection, but at the cost of substantially wider prediction sets and lower automation rates. The conformal framework degrades gracefully rather than failing silently when the exchangeability assumption is weakened. For a clinical informatics audience, this distinction matters: the conformal framework makes its assumptions and guarantees explicit, whereas the disagreement baseline requires the reader to trust an empirical calibration that may not transfer across institutions or time periods.

From a clinical workflow perspective, the Alert/Defer/Clear trichotomy maps naturally onto ICU triage operations. The Alert channel identifies patients for whom the model has high confidence of mortality risk, supporting early escalation decisions such as intensivist review, goals-of-care conversations, or preemptive resource allocation. At the union consensus operating point ($\alpha = 0.05$), alert PPV of 0.729 approaches the threshold at which automated alerts become actionable rather than noise-generating; approximately three out of four alerted patients are true positives. The Clear channel (used here as a model-confidence label indicating lower predicted risk, not as a clinical clearance for discharge or de-escalation) identifies patients for whom the model has high confidence of survival, supporting resource deprioritization or step-down consideration. Clear NPV of 0.962 means that fewer than 4% of cleared patients will experience the outcome, which is a clinically meaningful safety margin. The Defer channel, which absorbs the majority of cases at conservative $\alpha$ levels, mandates clinician review without model-driven recommendation. This is a deliberate design property: it prevents the model from making confident recommendations in the ambiguous middle of the risk distribution where discrimination is weakest. The decision-curve analysis reinforces this interpretation. At threshold 0.20, the continuous XGBoost score achieves net benefit 0.133 versus 0.082 for APACHE-III and 0.082 for SOFA, confirming that the ML model adds clinical value over existing severity scores.Vickers and Elkin [2006] It should be acknowledged that SOFA and APACHE-III were evaluated as single-feature logistic regression baselines, which is not how these scores are used in routine clinical practice. SOFA was designed as a monitoring tool for tracking organ dysfunction trajectories over time, and APACHE-III was designed as an admission severity score for case-mix adjustment and benchmarking rather than for individual patient triage. The comparison is therefore intended to establish a discrimination floor relative to commonly available clinical information, not to suggest that these scores are being used in their intended capacity. The conformal and disagreement policies should be read as complementary action-governance layers that sacrifice some net benefit for narrower, higher-confidence action sets.

The shift robustness findings have direct implications for multi-site deployment. Under simulated distribution shift (random missingness injection at 10–30% severity, measurement-process dropout, and care-process dropout), conformal coverage remained stable (0.947–0.970 at $\alpha = 0.05$) because the framework automatically deferred more cases as input quality degraded. This self-correcting behavior is a structural property of conformal prediction: when the nonconformity scores shift, the prediction sets widen, and more cases fall into the Defer region. By contrast, fixed-threshold policies experienced silent degradation. At 30% physiology missingness, the fixed 0.50 threshold retained precision of 0.619 but recall collapsed to 0.171, meaning the model missed 83% of true positives without any explicit signal that its operating characteristics had changed. This distinction is critical for deployment across institutions where documentation practices, measurement protocols, and care processes differ from the training environment. A conformal system that defers more when uncertain is safer than a fixed-threshold system that continues to make confident-looking predictions on degraded inputs. It should be noted that the simulated missingness perturbations follow a missing-completely-at-random (MCAR) mechanism, whereas the missingness analysis conducted on the training partition rejected the MCAR null hypothesis via Little's test. Real-world distribution shift between institutions is more likely to produce missing-not-at-random (MNAR) patterns, in which the probability of a measurement being absent depends on the unobserved value itself or on the patient's clinical trajectory. The MCAR simulation therefore represents a favourable scenario for shift robustness, and the framework's resilience to more realistic MNAR perturbations remains to be established.

The subgroup heterogeneity analysis reveals important patterns for deployment governance. At $\alpha = 0.10$, gender parity was preserved: female patients achieved coverage 0.921 with alert PPV 0.620, and male patients achieved coverage 0.902 with alert PPV 0.564, a gap within the pre-specified $\Delta \leq 0.05$ fairness bound for AUROC parity. Age-band stratification showed expected gradients, with younger patients ($\leq 50$) achieving the highest coverage (0.930) and certain-decision fraction (0.602), while the oldest stratum ($> 80$) had lower certain-decision fraction (0.344) reflecting greater clinical heterogeneity. The most clinically significant heterogeneity appeared across SOFA bands. Low-SOFA patients achieved coverage 0.949 with a small alert burden (3.7%), consistent with a population where mortality risk is genuinely low and the model can safely clear most cases. High-SOFA patients achieved lower coverage (0.866) with a much larger alert burden (31.6%) and alert PPV of 0.659, reflecting a sicker population where the model correctly identifies more patients as high-risk but also defers more cases. This SOFA-stratified pattern suggests a natural deployment governance policy: broader automation for low-acuity strata where the model is most reliable, and mandatory clinician oversight for high-acuity strata where the defer region is largest. Vasopressor use showed a similar gradient, with vasopressor-positive patients exhibiting lower coverage (0.885) and higher defer rates. These findings do not invalidate the framework; they define the boundaries within which it can be deployed responsibly.

# 5 Limitations

This work has several important limitations that bound the scope of its claims.

1. **Internal validation only.** The study is validated on a single-center MIMIC-derived cohort originating primarily from Beth Israel Deaconess Medical Center (BIDMC). No external cohort from a different institution or health system is included, so all performance estimates reflect internal discrimination and calibration. Transportability to other ICU populations, EHR systems, or documentation cultures is not established.

2. **No temporal split.** MIMIC-IV does not expose defensible calendar timestamps at the granularity needed for a true temporal validation split. The subject-grouped random split used here controls for patient-level leakage but does not test whether the model's operating characteristics are stable across calendar time. Temporal drift in clinical practice, coding conventions, or patient demographics could degrade performance in ways that this evaluation design cannot detect.

3. **Repeated subjects.** The working cohort contains 9,002 unique subjects across 10,036 ICU-stay-level rows, with 1,034 repeated-subject rows that make stay-level random splitting inappropriate for deployment claims. The package therefore treats subject-grouped evaluation as a hard design constraint rather than an optional robustness check. Patients with multiple ICU stays may differ systematically from single-stay patients, and the grouped split does

not fully eliminate within-subject correlation in the calibration and test sets. Furthermore, within the conformal calibration partition, repeated stays from the same subject produce nonconformity scores that are not fully independent. While the exchangeability assumption underlying conformal prediction is weaker than independence, within-subject correlation could affect quantile estimation. A sensitivity analysis restricting the calibration set to one stay per subject was not conducted but represents a useful robustness check for future work.

4. **Process features may not transport.** The 162-feature engineering pipeline includes care-process and measurement-process features (e.g., frequency of laboratory draws, timing of interventions) that reflect institutional workflows at BIDMC. These features contribute to discrimination in this cohort but may not generalize to institutions with different staffing models, monitoring protocols, or EHR documentation practices. The feature ablation in Appendix B (Table 10) shows that the physiology-severity subset achieves AUROC 0.756, only modestly below the full model's 0.763, suggesting that the process-feature contribution is real but not dominant.

5. **Conformal guarantees are marginal, not conditional.** The finite-sample coverage guarantee provided by conformal prediction holds marginally under the exchangeability assumption (that is, averaged over the entire test population).Vovk et al. [2005] It does not guarantee conditional coverage within specific subgroups (e.g., high-SOFA patients, elderly patients, or specific ethnic groups). The subgroup analysis in Table 6 shows that coverage varies across strata, with high-SOFA patients achieving only 0.866 coverage at $\alpha = 0.10$ compared with 0.949 for low-SOFA patients. Deployment governance must account for this heterogeneity.

6. **Secondary horizon results are limited by low event counts.** The 48-hour mortality endpoint has prevalence 0.004 and produced AUROC 0.705 with no usable PPV $\geq 0.50$ operating point. The 7-day endpoint (prevalence 0.115, AUROC 0.732) retains low recall at the conservative precision target. These secondary horizons should be treated as sensitivity analyses rather than standalone deployment targets.

7. **No prospective evaluation or clinician behavior study.** The manuscript focuses entirely on retrospective decision support. It does not evaluate whether the Alert/Defer/Clear framework changes clinician behavior, improves patient outcomes, or integrates smoothly into existing ICU workflows. Prospective validation with clinician-in-the-loop evaluation is a necessary next step before any deployment claim can be made.

8. **Disagreement baseline is heuristic.** The disagreement-based selective-triage comparator depends on an empirically tuned agreement threshold and is included as a methodological comparison rather than as the recommended deployment strategy. Its operating characteristics are not guaranteed to transfer across cohorts or model families.

9. **Feature engineering is bounded by the T24 window.** All predictor features are extracted from the first 24 hours of ICU admission. Features from later time points (treatment response trajectories, serial organ failure trends, and evolving care-process markers) could substantially improve discrimination but would shift the prediction horizon and change the clinical use case. The T24 constraint is a deliberate design choice that prioritizes early triage over maximal discrimination.

10. **Calibration may drift without recalibration infrastructure.** The isotonic calibration applied to the selected XGBoost model was fitted on the grouped validation set from this

cohort. In a deployed setting, calibration would need periodic monitoring and recalibration as patient populations, treatment protocols, and documentation practices evolve. No recalibration infrastructure or drift-detection mechanism is proposed in this work.

11. **No comparison with alternative uncertainty quantification methods.** Conformal prediction was selected for its distribution-free finite-sample coverage guarantees and model-agnostic applicability, but alternative uncertainty quantification approaches (Monte Carlo dropout, deep ensembles, Bayesian logistic regression, and calibration-based selective classifiers using the Chow reject option) were not benchmarked. Whether these alternatives could achieve more favourable coverage–automation tradeoffs without formal guarantees remains an open question.

12. **No individual-level feature explanations.** The manuscript focuses on the decision-framework contribution and does not include individual-level feature-importance analysis (e.g., SHAP or LIME) for the underlying XGBoost model. Clinicians receiving an Alert or Clear recommendation would benefit from knowing which features contributed most to that classification. Feature-level explanations for conformal triage decisions are identified as a priority for future work.

# 6 Conclusion

In this SA-AKI cohort derived from MIMIC, the predictive ceiling for first-24-hour ICU mortality risk is bounded at approximately AUROC 0.77 under leak-free subject-grouped evaluation, and that empirical fact changes the scientific target. The most defensible contribution of this work is not a marginal improvement in discrimination but an uncertainty-aware triage framework that makes explicit when the model should act and when it should defer. The conventional benchmark comparison (XGBoost with isotonic calibration at AUROC 0.768, outperforming APACHE-III at 0.579 and SOFA at 0.646 on decision-curve net benefit) establishes that machine learning adds value over existing severity scores. But the more important finding is that this value can be delivered safely only when the model's uncertainty is quantified and translated into actionable clinical categories.

Conformal selective triage provides that translation. It yields clinically interpretable Alert/Defer/Clear decisions with finite-sample coverage guarantees, stronger reliability under conservative multi-model consensus (union alert PPV 0.729, clear NPV 0.962 at $\alpha = 0.05$), and more graceful degradation under simulated distribution shift than fixed-threshold policies. The framework does not claim to solve the SA-AKI mortality prediction problem; it claims to define a more realistic boundary around what first-24-hour retrospective models can responsibly deliver. For institutions considering ML-assisted ICU triage, the practical message is that the model should be deployed as a triage assistant with an explicit defer option, not as an autonomous classifier, and that deployment governance should be stratified by patient acuity.

This work does not claim bedside readiness. External validation on independent cohorts, temporal validation across calendar periods, and prospective clinician-in-the-loop evaluation are all necessary before the Alert/Defer/Clear framework can be recommended for clinical use. What this work does claim is that conformal selective triage is a principled and transparent mechanism for navigating the gap between retrospective model development and safe clinical deployment, and that the bounded-ceiling finding should temper expectations about what first-24-hour SA-AKI mortality models can achieve regardless of the modeling approach.

**Table 7:** Appendix Table B1. Repeated grouped benchmark summary for the main tree-family comparison.

| Model | AUROC | Brier | ECE | Recall @ PPV≥0.50 |
|---|---|---|---|---|
| catboost | 0.768 | 0.161 | 0.022 | 0.623 |
| lightgbm | 0.765 | 0.161 | 0.021 | 0.575 |
| xgboost | 0.768 | 0.161 | 0.017 | 0.608 |

# Appendices

## A Data and Cohort

The working cohort was derived from MIMIC-IV v3.1, a publicly available critical care database covering ICU admissions at Beth Israel Deaconess Medical Center.Johnson et al. [2023] SA-AKI was defined by the co-occurrence of sepsis (Sepsis-3 criteria: suspected infection evidenced by antibiotic administration and culture sampling within specified temporal windows, plus Sequential Organ Failure Assessment score $\geq 2$)Singer et al. [2016], Vincent et al. [1996] and acute kidney injury (KDIGO staging criteria based on serum creatinine elevation or urine output reduction).Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group [2012] The cohort construction pipeline applied the following exclusion criteria: age $< 18$ years, ICU stays shorter than 24 hours, and missing key outcome variables. One row in the final dataset represents one ICU-stay-level SA-AKI record scored at 24 hours after ICU admission (T24).

The audit pass identified 9,002 unique subjects among 10,036 working rows, with 1,034 repeated-subject rows that make stay-level random splitting inappropriate for deployment claims. The package therefore treats subject-grouped evaluation as a hard design constraint rather than an optional robustness check. Feature engineering produced 162 predictor variables from the first 24 hours of ICU admission, using nine statistical aggregations (first, last, median, interquartile range, range, delta, area under the curve, slope, and count) applied to time-series physiological and process measurements.

The explicit data contract for the journal package is intentionally conservative: one row equals one ICU-stay-level SA-AKI record scored at T24; all predictor features must be available by T24; post-T24 information is excluded from deployment claims even if it exists elsewhere in the repository; and bedside-readiness, transportability, and exact thesis-cohort equivalence are not claimed.

## B Extended Benchmarks

This appendix gathers the benchmark, ablation, and comparator tables that support the main narrative without overloading the primary Results section.

### B.1 Tree-Family Benchmark

Table 7 reports the repeated grouped benchmark summary for the three tree-model families. The narrow performance band (AUROC 0.765–0.768) confirms that model-family selection is not the primary driver of discrimination in this cohort.

**Table 8:** Appendix Table B2. Grouped Optuna summary for the LightGBM benchmark.

| Model | Trials | Validation AUROC | Validation recall @ PPV≥0.50 | Validation Brier | Validation ECE |
|---|---|---|---|---|---|
| lightgbm | 40 | 0.752 | 0.590 | 0.177 | 0.093 |

**Table 9:** Appendix Table B3. Top grouped Optuna LightGBM trials.

| Model | Trial | Objective | AUROC | Brier | Recall @ PPV≥0.50 | Trees | Learning rate | Leaves | Max depth | Min child samples |
|---|---|---|---|---|---|---|---|---|---|---|
| lightgbm | 17 | 0.882 | 0.752 | 0.177 | 0.590 | 603 | 0.132 | 52 | 10 | 61 |
| lightgbm | 34 | 0.881 | 0.753 | 0.173 | 0.580 | 627 | 0.095 | 60 | 8 | 24 |
| lightgbm | 33 | 0.878 | 0.753 | 0.176 | 0.569 | 605 | 0.085 | 60 | 9 | 50 |
| lightgbm | 29 | 0.874 | 0.754 | 0.175 | 0.552 | 814 | 0.109 | 60 | 8 | 25 |
| lightgbm | 24 | 0.872 | 0.748 | 0.179 | 0.569 | 607 | 0.196 | 50 | 10 | 10 |

**Table 10:** Appendix Table B4. Feature ablation of the selected model.

| feature_set | model_name | auroc | auprc | brier | ece | precision | recall | f1 | specificity | alert_rate | prevalence | tp | fp | tn | fn | calibration_intercept | calibration_slope | recall_at_ppv_050 | precision_at_ppv_050_threshold | threshold_ppv_050 | low_risk_coverage_npv_095 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| physiology_severity | xgboost | 0.756 | 0.539 | 0.163 | 0.011 | 0.490 | 0.531 | 0.510 | 0.801 | 0.287 | 0.265 | 282.000 | 293.000 | 1177.000 | 249.000 | -0.058 | 0.986 | 0.552 | 0.500 | 0.365 | 0.173 |
| full | xgboost | 0.763 | 0.546 | 0.161 | 0.016 | 0.493 | 0.550 | 0.520 | 0.796 | 0.296 | 0.265 | 292.000 | 300.000 | 1170.000 | 239.000 | -0.032 | 1.009 | 0.548 | 0.500 | 0.356 | 0.167 |
| physiology_plus_care | xgboost | 0.756 | 0.536 | 0.163 | 0.025 | 0.499 | 0.501 | 0.500 | 0.818 | 0.266 | 0.265 | 266.000 | 267.000 | 1203.000 | 265.000 | -0.069 | 0.982 | 0.494 | 0.508 | 0.389 | 0.097 |

**Table 11:** Appendix Table B5. Secondary horizon sensitivity analyses.

| label | prevalence | auroc | auprc | brier | ece | precision | recall | f1 | specificity | alert_rate | tp | fp | tn | fn | calibration_intercept | calibration_slope | recall_at_primary_ppv |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| death_within_48h | 0.004 | 0.705 | 0.013 | 0.004 | 0.000 | 0.000 | 0.000 | 0.000 | 1.000 | 0.000 | 0.000 | 0.000 | 2020.000 | 9.000 | -0.168 | 0.964 | NA |
| death_within_7d | 0.115 | 0.732 | 0.261 | 0.094 | 0.011 | 0.393 | 0.048 | 0.085 | 0.990 | 0.014 | 11.000 | 17.000 | 1753.000 | 219.000 | -0.160 | 0.891 | 0.060 |
| death_before_discharge | 0.265 | 0.763 | 0.546 | 0.161 | 0.016 | 0.493 | 0.550 | 0.520 | 0.796 | 0.296 | 292.000 | 300.000 | 1170.000 | 239.000 | -0.032 | 1.009 | 0.548 |

## B.2 Hyperparameter Optimization

Tables 8 and 9 summarize the grouped Optuna search for LightGBM. The 40-trial search improved validation recall at PPV $\geq$ 0.50 to 0.590, but the repeated grouped benchmark still left XGBoost, tuned LightGBM, and CatBoost in a narrow performance band. This result reinforces the bounded-ceiling interpretation: systematic hyperparameter optimization does not break through the discrimination ceiling imposed by the T24 feature window.

## B.3 Feature Ablation

Table 10 reports the feature ablation of the selected XGBoost model across three feature subsets: physiology-severity only, physiology plus care-process, and the full feature set. The physiology-severity subset achieves AUROC 0.756, only 0.007 below the full model, indicating that the process-feature contribution is real but modest. This has implications for transportability: institutions where process features differ substantially from BIDMC may still achieve reasonable discrimination using physiology-only inputs.

## B.4 Secondary Horizons

Table 11 reports sensitivity analyses for secondary prediction horizons. The 48-hour endpoint (prevalence 0.004) produced AUROC 0.705 with no usable PPV $\geq$ 0.50 operating point, confirming that very-short-horizon mortality is too rare for reliable prediction with this feature set. The 7-day endpoint (prevalence 0.115, AUROC 0.732) is more stable but retains low recall at the conservative precision target (0.060). These results are included for completeness but should not be interpreted as standalone deployment targets.

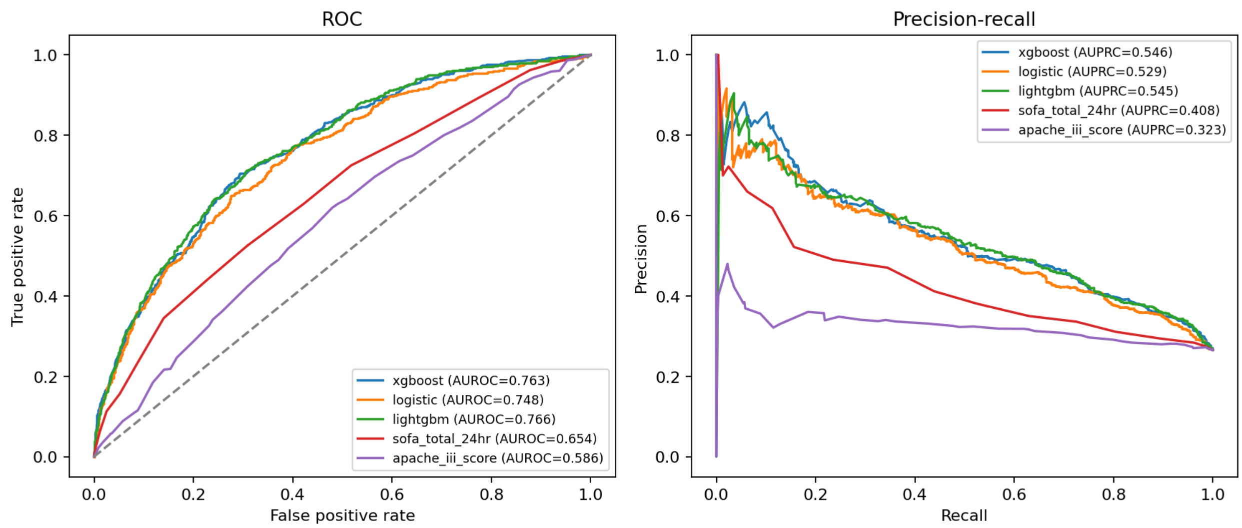


**Figure 10:** Appendix Figure B1. ROC and precision-recall benchmark curves.

**Table 12:** Appendix Table B6. Repeated grouped disagreement-based selective-triage summary.

| Policy | Alert precision mean | Alert precision std | Alert recall mean | Alert recall std | Actionable coverage mean | Actionable coverage std | Low-risk NPV mean | Low-risk NPV std | Defer rate mean | Defer rate std | Actionable error mean | Actionable error std |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| fixed_threshold | 0.491 | 0.007 | 0.613 | 0.056 | 0.520 | 0.052 | 0.955 | 0.009 | 0.480 | 0.052 | 0.343 | 0.008 |
| selective_triage | 0.484 | 0.025 | 0.225 | 0.151 | 0.260 | 0.139 | 0.960 | 0.013 | 0.740 | 0.139 | 0.254 | 0.033 |

**Table 13:** Appendix Table B7. Disagreement-based shift sensitivity summary.

| scenario | severity | policy_name | alert_rate | low_risk_coverage | defer_rate | actionable_coverage | alert_precision | alert_recall | low_risk_npv | defer_event_rate | actionable_error_rate | high_count | low_count | defer_count |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| physiology_missingness_shift | 0.100 | fixed_threshold | 0.278 | 0.164 | 0.557 | 0.443 | 0.503 | 0.527 | 0.936 | 0.206 | 0.336 | 557.000 | 329.000 | 1115.000 |
| physiology_missingness_shift | 0.100 | selective_triage | 0.093 | 0.128 | 0.779 | 0.221 | 0.543 | 0.190 | 0.926 | 0.264 | 0.235 | 186.000 | 256.000 | 1559.000 |
| measurement_process_dropout | 1.000 | fixed_threshold | 0.307 | 0.132 | 0.561 | 0.439 | 0.486 | 0.563 | 0.955 | 0.196 | 0.373 | 615.000 | 264.000 | 1122.000 |
| measurement_process_dropout | 1.000 | selective_triage | 0.100 | 0.100 | 0.800 | 0.200 | 0.475 | 0.179 | 0.955 | 0.267 | 0.285 | 200.000 | 200.000 | 1601.000 |
| care_process_dropout | 1.000 | fixed_threshold | 0.296 | 0.167 | 0.537 | 0.463 | 0.493 | 0.550 | 0.952 | 0.208 | 0.341 | 592.000 | 335.000 | 1074.000 |
| care_process_dropout | 1.000 | selective_triage | 0.106 | 0.133 | 0.761 | 0.239 | 0.514 | 0.205 | 0.951 | 0.269 | 0.243 | 212.000 | 266.000 | 1523.000 |

## B.5 ROC, Calibration, and Decision Curves

Figures 10–12 provide the full ROC and precision-recall curves, calibration curve for the deployed baseline, and decision curve for benchmark and fixed-policy comparators. The ROC and PR curves confirm the narrow discrimination band across model families. The calibration curve shows that isotonic recalibration produces well-calibrated probabilities across the risk spectrum. The decision curve demonstrates that the continuous XGBoost score dominates APACHE-III and SOFA across the clinically relevant threshold range (0.10–0.30).

## B.6 Disagreement-Based Selective Triage

Tables 12 and 13 summarize the disagreement-based selective-triage comparator. Table 12 shows the repeated grouped summary: the selective-triage policy reduced actionable error rate from 0.343 (fixed threshold) to 0.254, but at the cost of a much higher defer rate (0.740 versus 0.480). Table 13 reports the disagreement-based shift sensitivity, showing that the selective policy maintains lower actionable error under shift but with substantially reduced alert recall compared with conformal triage.

## B.7 Clinical Score Comparators

Tables 14 and 15 report the clinical score baselines. SOFA achieved repeated grouped AUROC 0.646 and APACHE-III achieved 0.579, both substantially below the ML baselines. At the PPV $\geq 0.50$

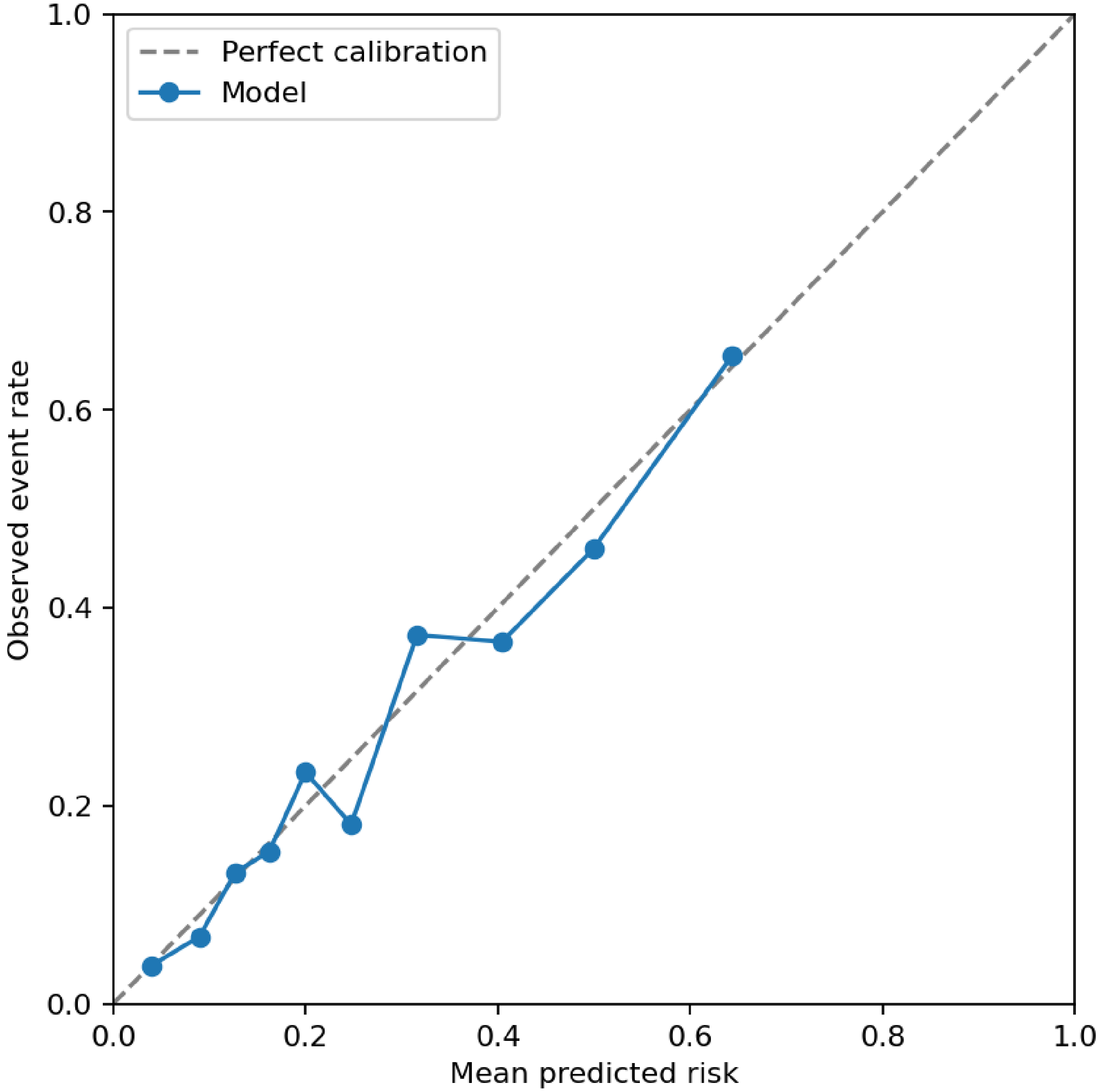


**Figure 11:** Appendix Figure B2. Calibration curve for the deployed baseline (XGBoost with isotonic calibration).

**Table 14:** Appendix Table B8. Repeated grouped clinical score summary.

| model_name | auroc__mean | auroc__std | auprc__mean | auprc__std | brier__mean | brier__std | ece__mean | ece__std | recall_at_ppv_050__mean | recall_at_ppv_050__std |
|---|---|---|---|---|---|---|---|---|---|---|
| apache_iii_score | 0.579 | 0.007 | 0.330 | 0.010 | 0.245 | 0.001 | 0.226 | 0.003 | 0.023 | 0.019 |
| sofa_total_24hr | 0.646 | 0.009 | 0.397 | 0.012 | 0.233 | 0.001 | 0.217 | 0.005 | 0.146 | 0.002 |

operating point, SOFA achieved recall 0.147 and APACHE-III achieved recall 0.023, confirming that single severity scores are insufficient for reliable mortality triage in this cohort. These scores remain useful as clinical anchors for decision-curve comparison but should not be considered competitive triage tools on their own.

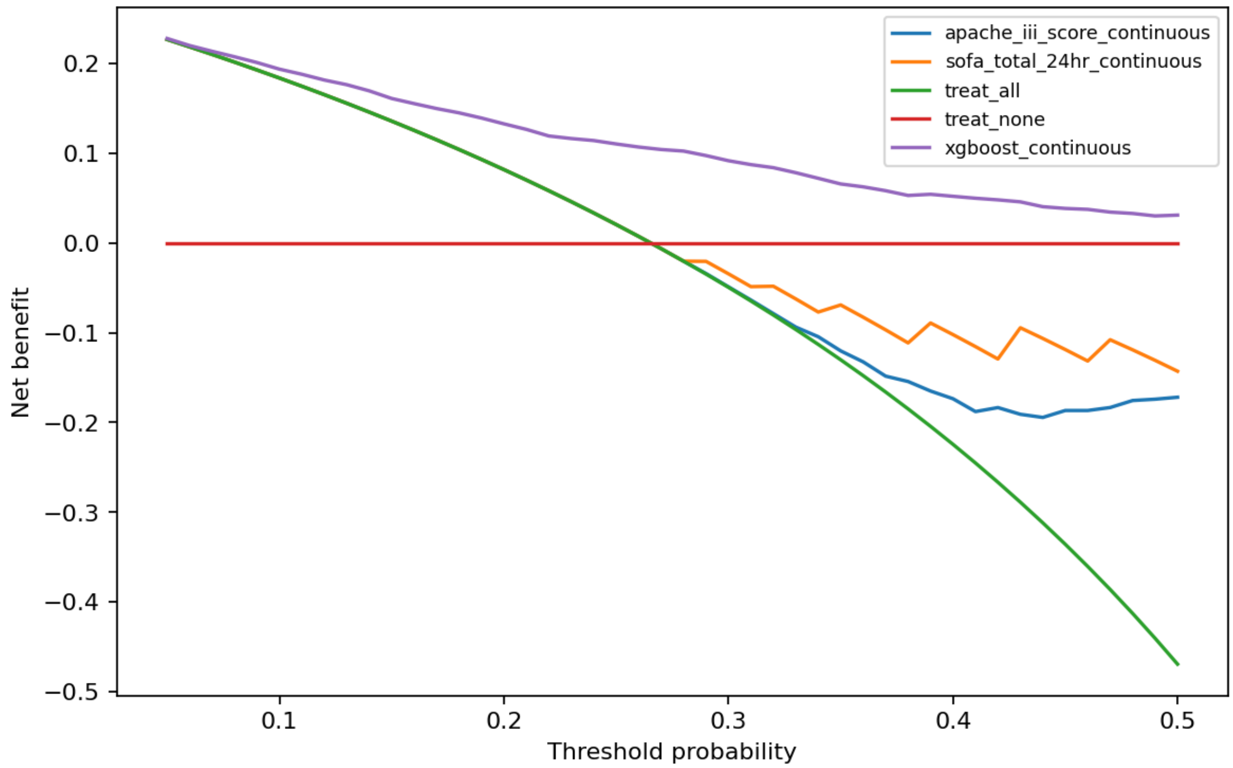


**Figure 12:** Appendix Figure B3. Decision curve for benchmark models and fixed-policy comparators.

**Table 15:** Appendix Table B9. Clinical score operating points on the main grouped split.

| score_name | auroc | auprc | brier | ece | precision | recall | f1 | specificity | alert_rate | prevalence | tp | fp | tn | fn | calibration_intercept | calibration_slope | threshold_ppv_050 | precision_at_ppv_050_threshold | recall_at_ppv_050_threshold | alert_rate_at_ppv_050_threshold |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| sofa_total_24hr | 0.654 | 0.408 | 0.232 | 0.221 | 0.619 | 0.113 | 0.191 | 0.975 | 0.048 | 0.265 | 60.000 | 37.000 | 1433.000 | 471.000 | -1.029 | 1.098 | 0.695 | 0.500 | 0.147 | 0.078 |
| apache_iii_score | 0.586 | 0.323 | 0.244 | 0.228 | 0.480 | 0.023 | 0.043 | 0.991 | 0.012 | 0.265 | 12.000 | 13.000 | 1457.000 | 519.000 | -1.014 | 0.931 | 0.658 | 0.571 | 0.028 | 0.013 |

## B.8 Clinical Utility

Tables 16 and 17 provide the full decision-curve net-benefit comparisons and fixed-policy utility summaries. At threshold 0.20, the continuous XGBoost score achieves net benefit 0.133, compared with 0.108 for the fixed PPV $\geq 0.50$ threshold, 0.059 for conformal $\alpha = 0.05$, and 0.035 for union conformal $\alpha = 0.05$. The conformal and union policies trade net benefit for higher alert precision and explicit deferral, which is the intended design: they are action-governance layers, not replacements for continuous risk ranking.

## B.9 Negative and Null Probes

Table 18 reports negative and null results from exploratory probe experiments, carried forward for transparency. Missingness-indicator features (AUROC 0.755), engineered ratio features (0.751), a three-model stacking ensemble (0.759), and HPO-tuned LightGBM (0.760) all failed to break through the discrimination ceiling. A random survival forest achieved concordance 0.726, below the primary XGBoost baseline. These null results reinforce the bounded-ceiling interpretation and justify the shift toward uncertainty quantification rather than continued architecture search.

**Table 16:** Appendix Table B10. Key decision-curve net-benefit comparisons.

| Threshold | Strategy | Net benefit | Standardized net benefit | Alert rate |
|---|---|---|---|---|
| 0.100 | apache_iii_score_continuous | 0.184 | 0.692 | 1.000 |
| 0.100 | conformal_alpha_0.05 | 0.064 | 0.242 | 0.107 |
| 0.100 | disagreement_selective | 0.045 | 0.170 | 0.099 |
| 0.100 | fixed_threshold_ppv50 | 0.129 | 0.487 | 0.296 |
| 0.100 | sofa_total_24hr_continuous | 0.184 | 0.692 | 1.000 |
| 0.100 | union_conformal_alpha_0.05 | 0.037 | 0.138 | 0.050 |
| 0.100 | xgboost_continuous | 0.194 | 0.730 | 0.821 |
| 0.200 | apache_iii_score_continuous | 0.082 | 0.308 | 1.000 |
| 0.200 | conformal_alpha_0.05 | 0.059 | 0.222 | 0.107 |
| 0.200 | disagreement_selective | 0.038 | 0.145 | 0.099 |
| 0.200 | fixed_threshold_ppv50 | 0.108 | 0.409 | 0.296 |
| 0.200 | sofa_total_24hr_continuous | 0.082 | 0.308 | 1.000 |
| 0.200 | union_conformal_alpha_0.05 | 0.035 | 0.131 | 0.050 |
| 0.200 | xgboost_continuous | 0.133 | 0.500 | 0.549 |
| 0.300 | apache_iii_score_continuous | -0.049 | -0.184 | 0.997 |
| 0.300 | conformal_alpha_0.05 | 0.052 | 0.196 | 0.107 |
| 0.300 | disagreement_selective | 0.030 | 0.112 | 0.099 |
| 0.300 | fixed_threshold_ppv50 | 0.082 | 0.308 | 0.296 |
| 0.300 | sofa_total_24hr_continuous | -0.034 | -0.130 | 0.952 |
| 0.300 | union_conformal_alpha_0.05 | 0.033 | 0.123 | 0.050 |
| 0.300 | xgboost_continuous | 0.092 | 0.345 | 0.364 |

**Table 17:** Appendix Table B11. Fixed-policy clinical utility summary.

| Strategy | Alert rate | Alert precision | Alert recall | Net benefit @0.10 | Net benefit @0.20 | Net benefit @0.30 |
|---|---|---|---|---|---|---|
| treat_all | 1.000 | 0.265 | 1.000 | 0.184 | 0.082 | -0.049 |
| treat_none | 0.000 | NA | 0.000 | 0.000 | 0.000 | 0.000 |
| fixed_threshold_ppv50 | 0.296 | 0.493 | 0.550 | 0.129 | 0.108 | 0.082 |
| disagreement_selective | 0.099 | 0.510 | 0.190 | 0.045 | 0.038 | 0.030 |
| conformal_alpha_0.05 | 0.107 | 0.640 | 0.258 | 0.064 | 0.059 | 0.052 |
| conformal_alpha_0.10 | 0.157 | 0.589 | 0.348 | 0.085 | 0.076 | 0.065 |
| union_conformal_alpha_0.05 | 0.050 | 0.752 | 0.143 | 0.037 | 0.035 | 0.033 |

**Table 18:** Appendix Table B12. Negative and null probe results carried forward for transparency.

| Probe | AUROC | AUPRC | Recall @ PPV≥0.50 |
|---|---|---|---|
| feat_missingness | 0.755 | 0.537 | 0.563 |
| feat_ratios | 0.751 | 0.526 | 0.514 |
| ensemble_stack_3 | 0.759 | 0.537 | 0.544 |
| hpo_lgbm | 0.760 | 0.540 | 0.595 |
| survival_rsf | 0.726 | NA | NA |

# C Shift and Subgroup Notes

The shift and subgroup results are intentionally framed as deployment governance notes rather than as new performance claims. Three shift scenarios were evaluated: random physiology missingness injection (10–30% severity), measurement-process feature dropout (complete removal of measurement-frequency features), and care-process feature dropout (complete removal of care-process features).

Under all shift scenarios, conformal coverage at $\alpha = 0.05$ remained between 0.947 and 0.970, compared with the clean baseline of 0.950. The mechanism is structural: as input quality degrades, nonconformity scores shift, prediction sets widen, and more cases enter the Defer region. At 30% physiology missingness, the conformal certain-decision fraction dropped from 0.316 (clean) to 0.235, meaning the system automatically became more conservative. Fixed-threshold policies, by contrast, experienced silent recall degradation from 0.294 (clean) to 0.171 at 30% missingness without any explicit signal to the clinician that the model's reliability had changed.

Subgroup analyses reinforce a governance-oriented interpretation. Low-acuity strata (low SOFA, no vasopressors, younger age) support broader automation with higher coverage and lower defer rates. High-acuity strata (high SOFA, vasopressor use, advanced age) require larger defer regions to maintain reliability. These patterns define where a cautious deployment policy should be most conservative and where automation can be extended with greater confidence. Gender parity was preserved across all operating points, with coverage differences between female and male patients remaining within 0.02.

# D Reproducibility

The manuscript and all associated tables, figures, and artifact summaries are generated from a fixed artifact bundle using a deterministic three-step build pipeline:

1. The deployment analysis script is executed to generate the canonical artifact bundle. This script performs the full evaluation pipeline including subject-grouped splitting, model training, calibration selection, conformal prediction, shift simulation, subgroup analysis, and decision-curve computation.

2. The manuscript build script materializes the chaptered markdown manuscript from the artifact bundle.

3. The LaTeX build script copies the official OUP template files, regenerates the figure and table assets, and writes the main document, bibliography, and package README into the manuscript directory.

The selected model snapshot preserved in the artifact bundle is XGBoost with isotonic calibration: mean AUROC 0.768 (cluster-bootstrap 95% CI: 0.741–0.787), mean AUPRC 0.544, mean recall at PPV $\geq$ 0.50 of 0.608, and decision-curve net benefit 0.133 at threshold 0.20. Headline conformal estimates were repeated across 21 grouped seeds to assess stability. The base random seed for preprocessing operations (imputation, feature selection, scaling) is 42. The 21 evaluation seeds used for repeated grouped benchmarking and conformal headline estimates are a fixed sequence of distinct integers (listed in the source code) that ensure independent subject-grouped splits across repetitions. The train/test split ratio is 80/20 with stratification by outcome.

Confidence intervals reported in Table 3 are 2.5th and 97.5th percentile intervals computed across the 21 per-seed point estimates. The conformal coverage guarantee is a per-split property: each of the 21 independent experiments is expected to satisfy $\Pr(y \in C(x)) \geq 1 - \alpha$ individually. Across the 21 seeds at $\alpha = 0.05$, per-seed coverage ranged from a minimum of 0.938 to a maximum of 0.968, with 19 of 21 seeds achieving coverage at or above the nominal 0.95 target.

The source patient-level data are not redistributed with this repository. Reproducing the cohort from raw source requires independent credentialed access to MIMIC-IV via PhysioNet and the corresponding data-use approvals. The checked-in repository provides the analysis code, derived artifact summaries, and manuscript assets used to support the internal-validation claims in this package.

## References


Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019. doi: 10.1145/3292500.3330701.

Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

Sean M. Bagshaw, Carol George, and Rinaldo Bellomo. Early acute kidney injury and sepsis: a multicentre evaluation. *Critical Care*, 12(2):R47, 2008. doi: 10.1186/cc6863.

Jialin Chen, Dan Wu, Chuyu Wu, Yiling Li, and Jing Yan. Machine learning-based prediction of ICU mortality in sepsis-associated acute kidney injury patients using MIMIC-IV database with validation from eICU database. *arXiv preprint arXiv:2502.17978*, 2025.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000. doi: 10.1161/01.CIR.101.23.e215.

Haibo He and Edwardo A. Garcia. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009.

Eric A. J. Hoste, Sean M. Bagshaw, Rinaldo Bellomo, Cynthia M. Cely, Roos Colman, Dinna N. Cruz, Kyriakos Edipidis, Lui G. Forni, Charles D. Gomersall, Deepak Govil, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Medicine*, 41(8):1411–1423, 2015. doi: 10.1007/s00134-015-3934-7.

Yue Hu, Haiyan Liu, and Bin Du. Prediction of in-hospital mortality in patients with sepsis-associated acute kidney injury using a cox proportional hazards model. *Frontiers in Medicine*, 8: 741837, 2021.

Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-wei H. Lehman, Leo A. Celi, and Roger G. Mark. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023. doi: 10.1038/s41597-022-01899-x.

Jin Soo Kang and Hee Jin Yoon. Current challenges in adopting machine learning to critical care and emergency medicine. *Acute and Critical Care*, 38(1):1–9, 2023. doi: 10.4266/acc.2022.01255.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

John A. Kellum, Xiaoyan Wen, Mark P. de Caestecker, Udayan Y. Bhatt, Nisha Bhatt, and Nisha Bhatt. Sepsis-associated acute kidney injury: a problem deserving of new solutions. *Nephron*, 145 (4):443–448, 2021. doi: 10.1159/000514940.

Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group. KDIGO clinical practice guideline for acute kidney injury. *Kidney International Supplements*, 2(1):1–138, 2012. doi: 10.1038/kisup.2012.1.

William A. Knaus, Douglas P. Wagner, Elizabeth A. Draper, Jack E. Zimmerman, Marilyn Bergner, Patricia G. Bastos, Christopher A. Sirio, Donald J. Murphy, Terry Lotring, Anne Damiano, and Frank E. Harrell. The APACHE III prognostic system: risk prediction of hospital mortality for critically ill hospitalized adults. *Chest*, 100(6):1619–1636, 1991. doi: 10.1378/chest.100.6.1619.

Yufeng Li, Rong Huang, Yanjun Geng, Tianhang Li, Weibo Li, and Jiayi Wang. Machine learning algorithm to predict mortality in critically ill patients with sepsis-associated acute kidney injury. *Scientific Reports*, 13(1):4893, 2023. doi: 10.1038/s41598-023-32160-z.

Jason T. Poston and Jay L. Koyner. Sepsis associated acute kidney injury. *BMJ*, 364:k4891, 2019. doi: 10.1136/bmj.k4891.

Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Fatemeh Roknaldin et al. Machine learning-based prediction of mortality in sepsis-associated acute kidney injury. *BMC Nephrology*, 25:42, 2024.

Emily J. See, Keerthi Jayasinghe, Neil Glassford, Michael Bailey, David W. Johnson, Kevan R. Polkinghorne, Nigel D. Toussaint, and Rinaldo Bellomo. Long-term risk of adverse outcomes after acute kidney injury: a systematic review and meta-analysis of cohort studies using consensus definitions of exposure. *Kidney International*, 95(1):160–172, 2019. doi: 10.1016/j.kint.2018.08.036.

Mervyn Singer, Clifford S. Deutschman, Christopher Warren Seymour, Manu Shankar-Hari, Djillali Annane, Michael Bauer, Rinaldo Bellomo, Gordon R. Bernard, Jean-Daniel Chiche, Craig M. Coopersmith, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*, 315(8):801–810, 2016. doi: 10.1001/jama.2016.0287.

Ewout W. Steyerberg, Andrew J. Vickers, Nancy R. Cook, Thomas Gerds, Mithat Gonen, Nancy Obuchowski, Michael J. Pencina, and Michael W. Kattan. Assessing the performance of prediction models: a framework for some traditional and novel measures. *Epidemiology*, 21(1):128–138, 2010. doi: 10.1097/EDE.0b013e3181c30fb2.

Ben Van Calster, David J. McLernon, Maarten van Smeden, Georg Bottolo, Leonardo and"; Heinze, Sujit Ghosh, and Ewout W. "; Steyerberg. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*, 17:230, 2019. doi: 10.1186/s12916-019-1466-7.

Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006. doi: 10.1177/0272989X06295361.

Jean-Louis Vincent, Rui Moreno, Jukka Takala, Sheila Willatts, Arnaldo De Mendonça, Hans Bruining, Christian K. Reinhart, Peter M. Suter, and L. G. Thijs. The SOFA (sepsis-related organ failure assessment) score to describe organ dysfunction/failure. *Intensive Care Medicine*, 22(7):707–710, 1996. doi: 10.1007/BF01709751.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

Alexander Zarbock, Mitra K. Nadim, Peter Pickkers, Hernando Gomez, Samira Bell, Michael Joannidis, Kianoush Kashani, Jay L. Koyner, Neesh Pannu, Melanie Meersch, et al. Sepsis-associated acute kidney injury: consensus report of the 28th Acute Disease Quality Initiative workgroup. *Nature Reviews Nephrology*, 19(6):401–417, 2023. doi: 10.1038/s41581-023-00683-3.